# KODA: Kinematics–Observation Data Alignment for Humanoid Learning from Human Egocentric Demonstration

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Training humanoid whole-body vision–language–action (VLA) policies requires costly robot demonstrations with aligned robot-view observations and controller-compatible whole-body state–action labels. Egocentric human demonstrations offer scalable supervision but have misaligned views and sparse motion cues. Existing approaches either pretrain policies that still require robot data for adaptation or generate controller-specific commands that limit reuse. Conversion also requires robot-view observations aligned with the generated whole-body motion. We present KODA (Kinematics–Observation Data Alignment), which converts egocentric RGB demonstrations into whole-body VLA training data using a reusable canonical trajectory to guide robot-view synthesis and generate controller-specific whole-body state–action supervision. KODA first generates a simulator-checked whole-body trajectory using a sparse tracker with sampling-based command optimization. This generated trajectory directly determines the onboard camera poses used to synthesize corresponding robot-view observations. The motion, observations, and task instruction form a canonical record. For each deployment controller, KODA optimizes commands to track the stored motion and records the corresponding action labels and proprioceptive states. When switching controllers, KODA retains this canonical record and regenerates whole-body state–action supervision for co-training with matching robot demonstrations. We evaluate KODA on four Unitree G1 loco-manipulation tasks using two whole-body controllers, SONIC and HEFT. With robot data held fixed, macro-average success on conditions covered by human demonstrations but absent from robot data improves by 45.0 and 46.25 percentage points, respectively. Under both controllers, this metric shows an upward trend across the evaluated human-data budgets, suggesting potential for further scaling.

**Keywords:** Egocentric Human Data, Humanoid Loco-Manipulation

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## 1 Introduction

Humanoid robots combine locomotion and dexterous manipulation in a versatile platform for operating in environments designed for people. Realizing this potential requires policies that coordinate body posture, mobility, and bimanual manipulation across the whole body. Whole-body vision–language–action (VLA) models aim to learn such coordination from onboard observations and task instructions (Jiang et al., 2026; Hu et al., 2026). Training requires corresponding robot observations, whole-body states, and action supervision compatible with the deployment whole-body controller (WBC) interface. Collecting these demonstrations on the target humanoid is costly, limiting coverage across tasks, objects, and environments.

Human demonstration interfaces and egocentric datasets provide RGB images, camera poses, and sparse bilateral hand-position references at scale (Chi et al., 2024; Hoque et al., 2026; Punamiya et al., 2026).

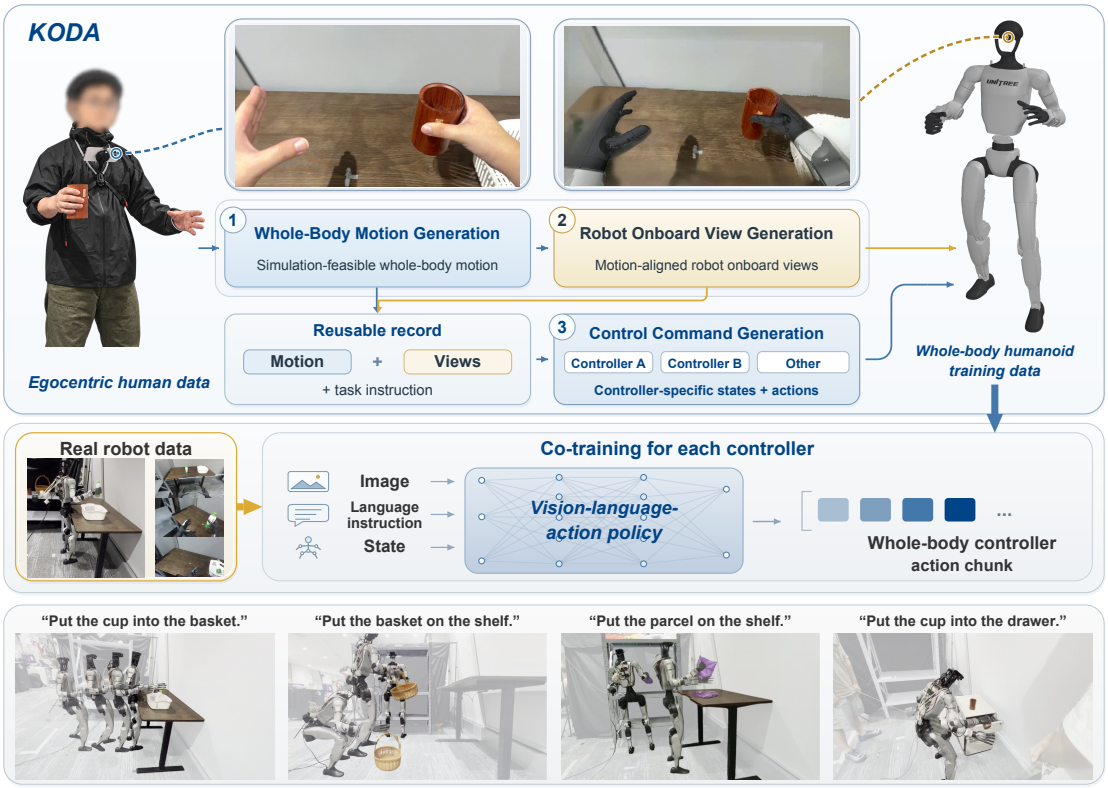



**Figure 1** Overview of KODA. **Whole-Body Motion Generation** produces simulation-feasible whole-body motion from human egocentric demonstrations, and **Robot Onboard View Generation** synthesizes motion-aligned onboard views. The motion, views, and task instruction form a reusable canonical record. **Control Command Generation** uses this record to generate controller-specific states and actions. The resulting data are combined with matching real-robot demonstrations to co-train a controller-specific vision–language–action (VLA) policy for language-guided humanoid loco-manipulation.

Their success on fixed-base arms (Chi et al., 2024) and in dexterous manipulation (Zheng et al., 2026b; Liu et al., 2026b) motivates extending human-data supervision to whole-body VLAs. Approaches such as EgoVLA, WholeBodyVLA, and MotionWAM learn human-derived priors for humanoid policies but still depend on target-humanoid demonstrations for adaptation (Yang et al., 2025b; Jiang et al., 2026; Zheng et al., 2026a). Target-humanoid data collection therefore remains a bottleneck. Other work converts human demonstrations into robot-oriented supervision. EgoHumanoid uses whole-body human measurements and a shared end-effector-and-navigation action space for co-training (Shi et al., 2026). HALOMI trains policies on human demonstrations using head–hand command labels tied to a dedicated controller (Zhao et al., 2026). These methods produce supervision tied to particular action interfaces, creating data silos that limit reuse across whole-body controllers; their outputs do not directly provide the whole-body states and compatible actions needed for other WBC interfaces. Their converted episodes also require observations corresponding to the generated robot motion. Fixed-offset reprojection (Shi et al., 2026) shifts the human camera without following the generated robot trajectory, leaving this observation–motion correspondence unresolved. Moreover, retaining only controller-tuned commands limits reuse to that controller's command space and tracking behavior. A reusable record must support adaptation to different deployment controllers.

We target a distinct data-conversion setting: augmenting an existing controller-specific humanoid whole-body dataset from egocentric RGB, camera poses, and hand references. This setting requires three capabilities: (i) generating simulator-checked whole-body trajectories from sparse human references; (ii) synthesizing onboard observations aligned with the generated robot motion; and (iii) instantiating the same converted record as compatible state–action supervision for different deployment controllers.

To address these three challenges, we present KODA (Kinematics–Observation Data Alignment), a framework

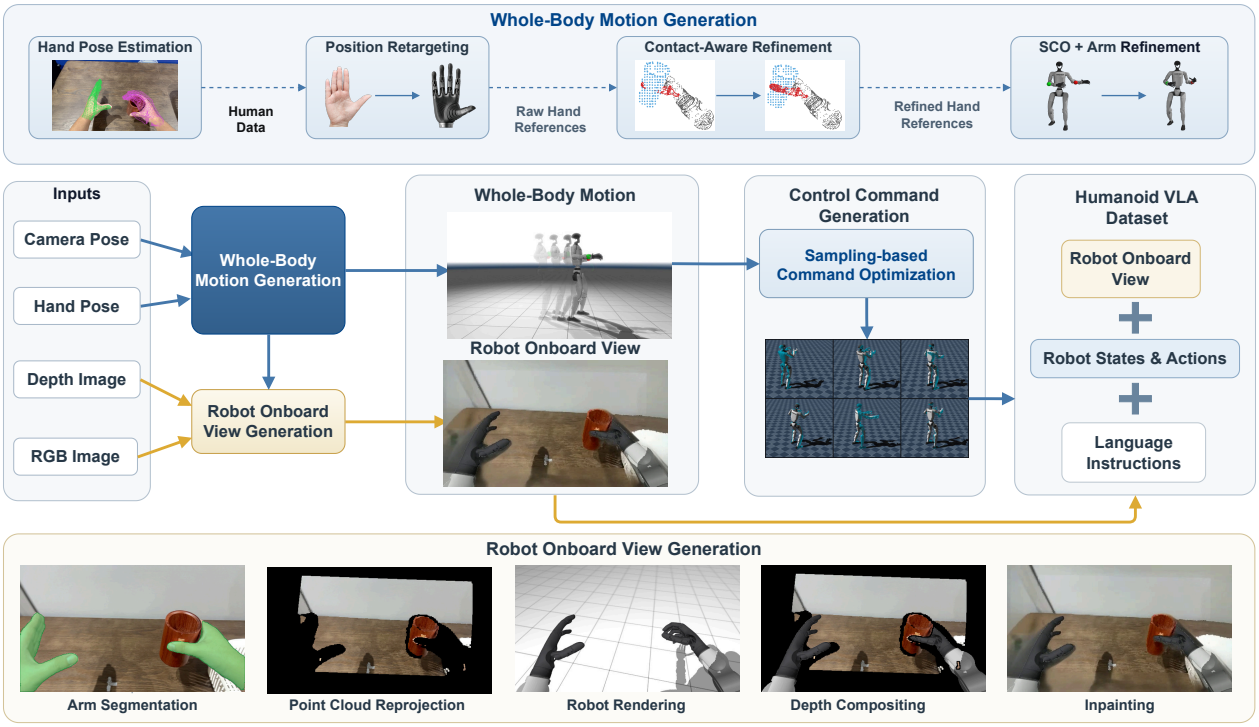



**Figure 2** KODA pipeline. **Middle:** Human demonstrations become aligned whole-body motion and onboard observations, then controller-specific state–action data for VLA training (Control Command Generation). **Top:** Whole-Body Motion Generation combines hand estimation and contact-aware retargeting, sampling-based command optimization (SCO), and arm refinement. **Bottom:** Robot Onboard View Generation combines arm removal, scene reprojection, robot rendering, depth compositing, and inpainting. Input depth is observed or estimated from RGB.

that converts egocentric RGB demonstrations into whole-body VLA training data. The conversion proceeds in three stages, as shown in Fig. 1. First, KODA generates a whole-body trajectory from sparse human references using a fine-tuned tracker. Sampling-based command optimization improves reference tracking while maintaining trajectory feasibility in simulation. Second, KODA derives camera poses from the generated trajectory and synthesizes corresponding onboard observations. The aligned motion and observations, together with the task instruction, form a canonical record for the target humanoid. Third, KODA optimizes the selected deployment controller's commands to track the fixed canonical trajectory, producing compatible state–action supervision for co-training with matching real-robot demonstrations. Changing controllers regenerates the state–action supervision while retaining the canonical motion and robot-view observations.

Our contributions are: (a) A data-conversion framework that constructs whole-body robot motion and corresponding onboard observations from egocentric human demonstrations. (b) A canonical representation and controller-specific control command generation procedure for generating compatible state–action supervision from the same converted records across deployment interfaces. (c) Real-world evidence of the training value and scaling potential of converted human demonstrations on four G1 loco-manipulation tasks. With 50 robot demonstrations per task fixed, adding 100 converted human demonstrations per task increases macro-average success by 45.0 and 46.25 percentage points under SONIC and HEFT, respectively, on object conditions covered by human training clips but absent from robot demonstrations. This metric shows an upward trend across the evaluated data budgets under both interfaces.

## 2 Related Work

### 2.1 Humanoid Loco-Manipulation

Humanoid data collection methods use direct teleoperation, human imitation, or whole-body tracking to obtain robot demonstrations. OmniH2O, HOMIE, HumanPlus, and TWIST2 provide interfaces for whole-

body teleoperation or imitation learning (He et al., 2024; Ben et al., 2025; Fu et al., 2024; Ze et al., 2025). OmniRetarget preserves human–object and human–environment interactions during retargeting, while SONIC tracks whole-body motion for physically grounded control (Yang et al., 2025a; Luo et al., 2026; Xie et al., 2025; Han et al., 2026). WholeBodyVLA learns loco-manipulation from vision and instructions, while OpenHLM and MotionWAM study whole-body humanoid policy and action-model training (Jiang et al., 2026; Hu et al., 2026; Zheng et al., 2026a; Xie et al., 2026b). These approaches establish the control and learning foundation for humanoid loco-manipulation, but their supervision is collected or represented in a target-robot interface.

### 2.2 Egocentric Data and Cross-Embodiment Alignment

Egocentric interfaces and datasets provide a scalable source of visual and motion supervision. UMI records robot-compatible gripper trajectories and images, while EgoDex and EgoVerse provide large-scale RGB data with camera and hand-motion annotations (Chi et al., 2024; Hoque et al., 2026; Punamiya et al., 2026). $\Psi_0$ and EgoVLA use human data for embodied pretraining, and EMMA, EgoEngine, and Ego2Robot combine human video with robot learning or robot-data synthesis (Wei et al., 2026; Yang et al., 2025b; Zhu et al., 2026; Liu et al., 2026b; Wang et al., 2026). Related cross-embodiment methods address appearance and viewpoint transfer through cross-painting or robot-view augmentation (Chen et al., 2024a,b).

For humanoids, EgoHumanoid uses whole-body measurements, fixed-offset view alignment, and training-time perturbations (Shi et al., 2026). HALOMI adopts this view alignment and reference adaptation based on the cross-entropy method (CEM) for a dedicated controller (Zhao et al., 2026). KODA also uses command optimization, but derives onboard views from generated whole-body motion. Its canonical motion–observation records can be reused to augment manipulation datasets for compatible WBCs through controller-specific command generation and state recording.

## 3 System Framework

KODA converts egocentric human episodes into robot-view observations, canonical whole-body motion, and controller-compatible state–action data for VLA training as shown in Fig. 2. The conversion has three stages: Whole-Body Motion Generation in Sec. 3.2 produces a simulator-checked trajectory from sparse human references; Robot Onboard View Generation in Sec. 3.3 uses the corresponding body–camera trajectory to synthesize aligned observations, forming a controller-independent canonical record; and Control Command Generation in Sec. 3.4 tracks this motion with the selected controller to derive its state–action supervision. The canonical motion and observations are reused across interfaces, while controller-specific states and commands enable co-training with matching real-robot demonstrations. Qualitative demonstration timelines are provided in Appendix E.

### 3.1 Problem Setup

Let $\mathcal{R}$ denote the target humanoid and $\kappa$ a deployment controller with its action interface. All sequences use the target control clock $t = 1, \dots, T$.

After geometry recovery, an RGB episode is represented as $e_h = (\{I_t^h, D_t^h, C_t^h, J_t^h\}_{t=1}^T, \ell)$, where $I_t^h$ is the egocentric RGB image, $D_t^h$ is observed or estimated depth, $C_t^h$ is the source-camera pose, $J_t^h$ contains the bilateral 3-D hand keypoints, and $\ell$ is the task instruction. Camera poses and hand keypoints come from source annotations or offline recovery. When source depth is unavailable, KODA estimates it from RGB using the metric-depth model of Depth Anything 3 (Lin et al., 2025). Depth, camera poses, and hand references must share a consistent metric coordinate system. Human episodes form $\mathcal{D}_{\text{human}}$. Real-robot episodes in $\mathcal{D}_{\text{robot}}^{\kappa}$ contain robot-view observations $I_t^{\mathcal{R}}$, controller-specific state features $s_t^{\kappa}$, actions $a_t^{\kappa}$, and instructions $\ell$.

$\mathcal{C}_{\mathcal{R}}$ constructs a canonical record in $\mathcal{R}$'s observation and kinematic spaces, without controller-specific states or actions: $e_{\mathcal{R}}^{\text{can}} = \mathcal{C}_{\mathcal{R}}(e_h) = \left(\left\{I_t^{\mathcal{R}}, q_t^{\mathcal{R}}\right\}_{t=1}^{T}, \ell\right)$. Here $q_t^{\mathcal{R}}$ is the whole-body configuration corresponding to $I_t^{\mathcal{R}}$, including the floating-base pose and body and hand joint positions. The canonical trajectory provides a common motion reference for robot-view synthesis and controller-specific tracking.

The mappings $\mathcal{C}_{\mathcal{R}}$ and $\mathcal{G}_{\kappa}$ act episode-wise on datasets:

$$\begin{aligned} e^{\kappa}_{\mathcal{R}} &= \mathcal{G}_{\kappa}(e^{\text{can}}_{\mathcal{R}}) = \left(\left\{I^{\mathcal{R}}_t, s^{\kappa}_t, a^{\kappa}_t\right\}_{t=1}^{T}, \ell\right), \\ \mathcal{D}^{\kappa}_{\text{KODA}} &= \mathcal{G}_{\kappa}(\mathcal{C}_{\mathcal{R}}(\mathcal{D}_{\text{human}})), \quad \mathcal{D}^{\kappa}_{\text{train}} = \mathcal{D}^{\kappa}_{\text{robot}} \cup \mathcal{D}^{\kappa}_{\text{KODA}}. \end{aligned} \tag{1}$$

The mapping $\mathcal{G}_{\kappa}$ retains the canonical observations and instruction, adding state features and optimized command labels in $\kappa$'s interface as described in Sec. 3.4. Converted and real-robot episodes share a format and jointly train the VLA policy with parameters $\theta$ to predict $\widehat{a}^{\kappa}_t = \pi_{\theta}(I^{\mathcal{R}}_t, s^{\kappa}_t, \ell)$.

### 3.2 Whole-Body Motion Generation

Given the camera poses and hand references in $e_h$, Whole-Body Motion Generation outputs the simulator-checked canonical whole-body motion $\{q^{\mathcal{R}}_t\}_{t=1}^{T}$.

*Contact-Aware Hand Retargeting* HAWOR (Zhang et al., 2025) estimates world-space MANO hand poses from egocentric RGB. Following AnyTeleop (Qin et al., 2023), KODA retargets them to obtain initial robot wrist poses and active finger-joint values using MANO palm geometry and a calibrated wrist-to-palm transform; distal mimic joints follow a fixed coupling map (Qin et al., 2023). Contact-aware refinement then adjusts this initialization on reliable contact frames. To identify these frames, EfficientSAM3 (Zeng et al., 2025) segments hand/forearm pixels, while nearby valid non-hand depths provide object-surface samples. Reliable contact frames are identified from multi-finger proximity, depth, and local surface consistency, then temporally filtered; without reliable contact, the initialization is retained.

At contact frames, wrist and finger poses are refined to preserve hand–object geometry. Following OmniRetarget (Yang et al., 2025a), selected human landmarks and corresponding robot-hand points are paired with the same nearby object samples to form connected world-frame interaction meshes. Each vertex descriptor is its offset from the uniformly weighted mean of its neighbors. We optimize wrist poses and finger joints to minimize differences between corresponding human and robot descriptors, preserving the demonstrated hand–object geometry. During optimization, object samples remain fixed, robot points are recomputed by forward kinematics, wrist translations, local rotations, and finger corrections are bounded, and deviations from initialization are penalized. Corrections are interpolated between contact frames and held constant outside their range. The optimized bilateral-wrist and finger trajectories are denoted by $w^{\text{ret}}_{1:T}$ and $f^{\text{ret}}_{1:T}$, respectively; the tracking stage uses $C^h_t$ and $w^{\text{ret}}_{1:T}$ as sparse inputs. See Appendix A.1 for details.

*Whole-body tracking via command optimization* The source-camera poses are converted to torso poses using the camera-to-torso transform from the target robot's URDF, $p^{\text{torso}}_t = \mathcal{T}_{\text{cam}\to\text{torso}}(C^h_t)$. Combining $p^{\text{torso}}_t$ with $w^{\text{ret}}_t$, concatenating the tracks in a fixed order, and applying a Savitzky–Golay filter (Schafer, 2011) yields $r^{\text{ref}}_{1:T}$.

Kimodo (Rempe et al., 2026) generates the initial whole-body configuration $q^{\mathcal{R}}_{\text{init}}$ from the first reference frame. After kinematic and physics checks, this configuration initializes the simulated rollout.

We add a three-point torso-and-bilateral-wrist tracking mode to ScaleBFM (Zeng et al., 2026) and fine-tune the pretrained whole-body controller $\phi_{\mathcal{R}}$ for the target embodiment. Its whole-body joint targets are applied by PD control in MuJoCo/MJX, but direct tracking can accumulate reference-tracking errors, as shown in TABLE 4. KODA therefore uses the cross-entropy method (CEM) (Pinneri et al., 2021) to optimize the sparse pose commands so that the realized motion tracks $r^{\text{ref}}_{1:T}$ more accurately while remaining simulation-feasible. Candidate commands are evaluated by simulated rollouts in a receding-horizon procedure detailed next and illustrated in Fig. 3; settings are reported in Sec. 4.1.

At each replan, $\mathbf{r}^{\text{ref}}$ spans $H$ steps from simulated state $x_{\text{cur}}$. The latent vector $z \in \mathbb{R}^d$ contains 18 translation/rotation-vector coordinates per knot for torso and bilateral-wrist SE(3) residuals. CEM initializes diagonal-Gaussian parameters $(\mu, \sigma) = (0, \sigma_0)$ and samples one residual for each of the $N$ parallel MJX environments per iteration:

$$\begin{aligned} z^{(n)} &= \mu + \sigma \odot \varepsilon^{(n)}, \qquad \varepsilon^{(n)}_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \\ \widetilde{\mathbf{r}}^{(n)} &= \text{Decode}(\mathbf{r}^{\text{ref}}, z^{(n)}), \end{aligned} \tag{2}$$

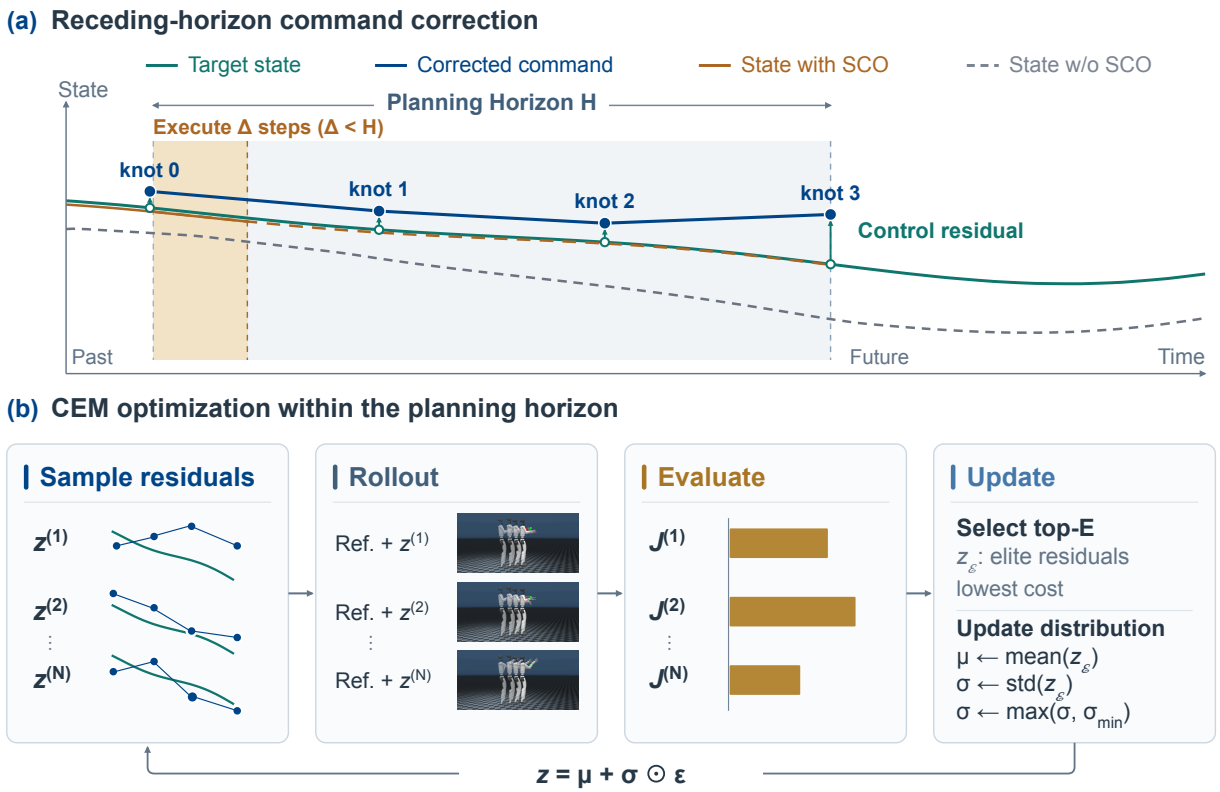



**Figure 3** Sampling-based command optimization (SCO) uses CEM to optimize sparse pose residuals through sampling, rollout, evaluation, and proposal updates. (a) Receding-horizon correction with a planning horizon and executed prefix. (b) The CEM loop proceeds from sampling to rollout, evaluation, and distribution update; the feedback arrow returns to sampling.

where $\odot$ denotes element-wise multiplication, $i = 1, \ldots, d$ indexes latent coordinates, and $n = 1, \ldots, N$ indexes candidates and environments. The operator Decode applies the sampled residuals at the reference knots, clips them to bounded physical ranges, and interpolates the translation and rotation corrections between knots to form the complete candidate pose-command trajectory $\widetilde{\mathbf{r}}^{(n)}$. We roll out the $N$ candidates in parallel MJX environments:

$$\widehat{\mathbf{q}}^{(n)} = \mathrm{Rollout}_{\mathrm{MJX}}(x_{\mathrm{cur}}, \phi_{\mathcal{R}}, \widetilde{\mathbf{r}}^{(n)}), \quad \widehat{\mathbf{r}}^{(n)} = \Pi_{\mathcal{B}}(\widehat{\mathbf{q}}^{(n)}), \tag{3}$$

Here $\widehat{\mathbf{q}}^{(n)}$ is the whole-body configuration trajectory realized by the $n$-th candidate command in MJX, and $\widehat{\mathbf{r}}^{(n)} = \Pi_{\mathcal{B}}(\widehat{\mathbf{q}}^{(n)})$ is the corresponding torso-and-wrist pose trajectory. The set $\mathcal{B}$ contains these torso and wrist bodies, and $\Pi_{\mathcal{B}}$ extracts their poses. Each realized trajectory is scored against the unperturbed reference by $J^{(n)} = d_{\mathcal{B}}(\widehat{\mathbf{r}}^{(n)}, \mathbf{r}^{\mathrm{ref}})$, a position/orientation tracking cost over these three links with receding-horizon time weights. Kinematic or physics violations are rejected. For the $E$ valid lowest-cost samples indexed by $\mathcal{E}$, we update each dimension with $\mu_i^+ = \mathrm{mean}_{n\in\mathcal{E}}\left[z_i^{(n)}\right]$ and $\sigma_i^+ = \max\{\sigma_{\min,i}, \mathrm{std}_{n\in\mathcal{E}}\left[z_i^{(n)}\right]\}$, where the standard deviation uses divisor $E$ and is bounded below by $\sigma_{\min,i}$. After $I$ updates $(\mu, \sigma) \leftarrow (\mu^+, \sigma^+)$, we select the lowest-cost valid rollout among the nominal, best sampled, and final-mean candidates. Only the first $\Delta$ steps are retained; the resulting simulator state initializes the next replan, whose horizon advances by $\Delta$. Concatenated prefixes form $q_{1:T}^{\mathcal{R},\mathrm{SCO}}$ for constrained refinement. See Appendix A.2 for implementation details.

*Constrained Arm Motion Refinement* To reduce residual wrist-tracking error, temporal Mink IK (Zakka, 2024) refines only the arm chain, including wrists. Contact-aware finger trajectories are retained; the optimized floating base, torso, and lower body remain fixed to preserve locomotion and balance motion. Refinement enforces joint limits, per-step displacement and velocity bounds, and self-collision constraints, while penalizing deviations from $q_{1:T}^{\mathcal{R},\mathrm{SCO}}$. Combining the refined arms and contact-aware fingers with the fixed non-arm

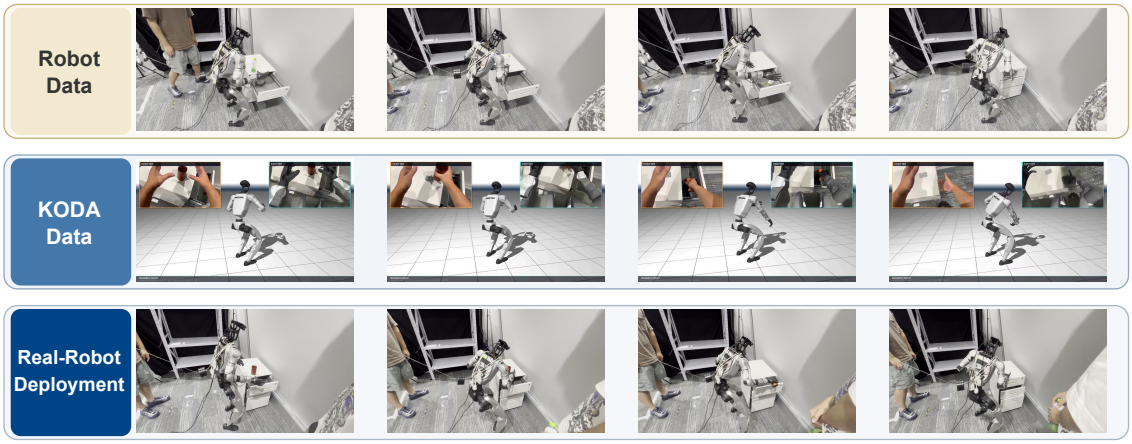



**Figure 4** Qualitative example of the task "Put the cup into the drawer." The rows show real-robot demonstrations (top), converted KODA data with aligned robot onboard observations (middle), and real-robot deployment of the learned VLA policy in the Robot-OOD setting (bottom). In the middle row, the main panels show the generated whole-body motion; the upper-left and upper-right insets show the source human view and the synthesized robot onboard view, respectively.

motion yields $q^{\mathcal{R}}_{1:T}$ for Robot Onboard View Generation. Replay shows comparable controller trackability after refinement, as quantified in Sec. 4.5. Further details are in Appendix A.3.

### 3.3 Robot Onboard View Generation

The optimized whole-body motion $q^{\mathcal{R}}_{1:T}$ from the previous section determines the robot camera trajectory. Because camera poses map camera coordinates to the world frame, the calibrated rigid torso–camera transform gives $C^{\mathcal{R}}_t = T_{\text{cam}}(q^{\mathcal{R}}_t)$. KODA then synthesizes $I^{\mathcal{R}}_t$ from the source RGB frame $I^h_t$, associated depth $D^h_t$, source-camera pose $C^h_t$, and target configuration $q^{\mathcal{R}}_t$; all stages use this target camera pose and timestamp.

EfficientSAM3 (Zeng et al., 2025) predicts an arm/hand mask $M^{\text{arm}}_t$; depth inside it is invalidated to obtain $\tilde{D}^h_t$. KODA back-projects remaining valid pixels using $\tilde{D}^h_t$, calibrated camera intrinsics, and $C^h_t$, transforms them to the target camera $C^{\mathcal{R}}_t$, and splats samples with a metric z-buffer, producing $(I^{\text{geo}}_t, D^{\text{geo}}_t, H^{\text{geo}}_t)$: the reprojected image, depth map, and hole mask. MuJoCo renders $(I^{\text{rob}}_t, D^{\text{rob}}_t)$ from the same $q^{\mathcal{R}}_t$ and camera. At pixels where both depth maps are valid, the nearer surface with smaller depth is retained: for pixel $u$, $M^{\text{rob}}_t(u) = 1$ if and only if $D^{\text{rob}}_t(u) < D^{\text{geo}}_t(u)$. If only one depth is valid, its corresponding pixel is retained; neighboring valid depths are used only inside geometric holes. The composite image is therefore $I^{\text{comp}}_t = M^{\text{rob}}_t I^{\text{rob}}_t + (1 - M^{\text{rob}}_t) I^{\text{geo}}_t$. DiffuEraser (Li et al., 2025) is applied only to residual holes $H^{\text{rem}}_t = H^{\text{geo}}_t \setminus M^{\text{rob}}_t$, namely, pixels left uncovered by both the reprojected scene and the rendered robot. Valid scene and robot pixels are frozen. The completed $I^{\mathcal{R}}_t$ is indexed on the target control clock and paired with $q^{\mathcal{R}}_t$. Together, the aligned observations $I^{\mathcal{R}}_{1:T}$, the whole-body motion $q^{\mathcal{R}}_{1:T}$ generated above, and the instruction $\ell$ form the canonical record $e^{\text{can}}_{\mathcal{R}}$, which the next section instantiates into controller-specific supervision. See Appendix B for details.

### 3.4 Control Command Generation and VLA Training

Given $e^{\text{can}}_{\mathcal{R}}$, KODA generates controller-specific command sequences for SONIC (Luo et al., 2026) and HEFT (Liu et al., 2026a) on the Unitree G1 with BrainCo hands. Our command generation is related to test-time command optimization (Xie et al., 2026a), which searches for controller-specific input commands through closed-loop simulation to improve motion tracking. At 50 Hz, SCO optimizes a shared pelvis-pose and 29-body-joint residual space (hands excluded) against fixed $q^{\mathcal{R}}_{1:T}$. The G1 reference contains world-frame pelvis position and quaternion, 29 body-joint positions and velocities, and world-frame link poses from forward kinematics. Each candidate is decoded by the selected controller and rolled out in simulation with the same cost for both controllers—joint-position MSE, 14-link pose error, reference deformation, PD-target change, and terminal tracking penalties—and the same Control Command Generation settings reported in Sec. 4.1.

**Table 1** Real-robot task success (%) with increasing KODA data. Each controller uses 50 robot demonstrations per task. KODA demos denotes the number of converted human demonstrations per task; zero denotes robot-only training. Avg. is the macro-average over four tasks. Bold marks the best result within each controller, including ties.

| Controller | KODA demos | Cup-to-basket | | Parcel-to-shelf | | Basket-to-shelf | | Cup-to-drawer | | Avg. | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD |
| SONIC | 0 | 55.00 | 0.00 | 90.00 | 15.00 | 55.00 | 0.00 | 50.00 | 0.00 | 62.50 | 3.75 |
| | 25 | 55.00 | 10.00 | **100.00** | 25.00 | 50.00 | 15.00 | **70.00** | 15.00 | 68.75 | 16.25 |
| | 50 | 65.00 | 20.00 | 90.00 | 65.00 | 60.00 | 10.00 | 65.00 | 0.00 | 70.00 | 23.75 |
| | 75 | **70.00** | 35.00 | 90.00 | 70.00 | **65.00** | 30.00 | 60.00 | 30.00 | 71.25 | 41.25 |
| | 100 | **70.00** | **40.00** | 90.00 | **75.00** | **65.00** | **45.00** | 65.00 | **35.00** | **72.50** | **48.75** |
| HEFT | 0 | 65.00 | 0.00 | 85.00 | 20.00 | 65.00 | 0.00 | 60.00 | 0.00 | 68.75 | 5.00 |
| | 25 | 65.00 | 5.00 | 90.00 | 35.00 | 45.00 | 20.00 | 65.00 | 20.00 | 66.25 | 20.00 |
| | 50 | 60.00 | 30.00 | 80.00 | 55.00 | 65.00 | 25.00 | **70.00** | 15.00 | 68.75 | 31.25 |
| | 75 | 70.00 | 40.00 | **95.00** | 65.00 | 60.00 | 35.00 | 65.00 | **40.00** | 72.50 | 45.00 |
| | 100 | **80.00** | **50.00** | 90.00 | **75.00** | **70.00** | **40.00** | 65.00 | **40.00** | **76.25** | **51.25** |

The full search and objective are specified in Appendices C.1 and C.2.

The resulting actions $a_t^\kappa$ use different controller interfaces. SONIC's motion encoder maps the optimized motion references to 64-dimensional motion tokens (Luo et al., 2026). HEFT uses 29 absolute body-joint targets and a 9-dimensional pelvis code, giving 38 body-reference dimensions. The HEFT pelvis code represents roll $r$ and pitch $p$ as $(\sin r, \cos r - 1, \sin p, \cos p - 1)$, together with a wrapped next-step yaw change, three translation increments in the yaw-aligned frame, and absolute height. Planar position and heading accumulate from an anchor, whereas height is read directly. Both controllers append 12 hand-joint targets, giving 76-dimensional SONIC and 50-dimensional HEFT actions. Pose retargeting alone does not specify grasp force. KODA thresholds each retargeted robot hand's mean finger curvature following EgoHumanoid (Shi et al., 2026). For grasp, hand-joint targets are set to a calibrated BrainCo grasp pose to encourage active closure; for release, targets equal the current retargeted joint angles. The 12 targets are sent separately to hand control.

Each converted episode records a 44-dimensional state vector $s_t^\kappa$ in a format shared by both controllers: 29 body-joint angles, three-dimensional rollout gravity rotated to the robot base frame by the base quaternion, and 12 continuous BrainCo hand-joint angles from contact-aware retargeting as geometric hand-state estimates. Canonical images and the instruction are retained; TABLE 4 reports wrist/camera pose residuals, and TABLE 1 reports co-training utility. GR00T N1.7 predicts 40-step chunks of $a_t^\kappa$ from $(I_t^{\mathcal{R}}, s_t^\kappa, \ell)$ using converted and real-robot episodes in the selected format. Changing the deployment interface reruns control command generation and state recording while reusing the canonical motion, observations, and instruction. Interface and export details are in Appendices C.3–C.5.

# 4 Evaluation

We evaluate whether KODA improves real-robot task success by augmenting fixed robot datasets with human demonstrations, reusing the same canonical records across SONIC and HEFT. EgoHumanoid (Shi et al., 2026) and HALOMI (Zhao et al., 2026) use different control interfaces and do not directly provide compatible whole-body state–action data. SONIC ablations assess the training benefits of observation alignment and controller-specific supervision; tracking diagnostics characterize motion fidelity and controller realization.

## 4.1 Experimental Setup

*Platform, data, and evaluation.* We use a Unitree G1 with BrainCo dexterous hands on four whole-body tasks: cup-to-basket, parcel-to-shelf, basket-to-shelf, and cup-to-drawer. Each task has 50 real-robot demonstrations from the corresponding SONIC or HEFT PICO teleoperation pipeline and 100 human egocentric demonstrations captured with a chest-mounted iPhone using MobileEgo Anywhere (Palanisamy et al., 2026). All 400 human clips include iPhone-recorded depth. Both sources use 50 Hz. The scaling study evaluates SONIC (Luo et al., 2026) and HEFT (Liu et al., 2026a); component ablations use SONIC with 50 robot and 100 converted-human demonstrations per task. Per controller configuration, GR00T N1.7 (Bjorck et al., 2025) trains on all four tasks for 20,000 iterations with batch size 160. Within each task, robot/human sampling is proportional to episode counts; other sampling settings follow GR00T defaults. We report one training run per configuration and 20

real-robot trials per task and condition. This trial count matches EgoHumanoid's 20 trials per setting (Shi et al., 2026) and exceeds the 10 trials per task and model reported in $\Psi_0$ (Wei et al., 2026). Success requires completing the placement without a fall or unrecovered manipulation failure. Reported averages weight the four tasks equally.

*Conditions covered by human demonstrations.* Robot-ID uses object appearance and pose conditions present in the robot demonstrations. Robot-OOD uses conditions absent from those demonstrations: a green bottle is replaced by a brown bottle in cup-to-basket and cup-to-drawer, a purple parcel by a white parcel in parcel-to-shelf, and the basket handle is tilted by approximately 30° in basket-to-shelf. The human training pool contains 80 shifted and 20 robot-matched clips per task. Robot-OOD tests whether human demonstrations supply coverage missing from the robot dataset, rather than zero-shot generalization to conditions absent from all training data. Data and evaluation details are in Appendices D.1 and D.2.

*CEM settings.* Both stages run at 50 Hz and use CEM with 80 candidates per iteration, 8 iterations per replan, and a 40-step horizon. Whole-Body Motion Generation uses 10 elites and executes 10 steps before replanning. Control Command Generation with SONIC and HEFT uses 9 elites and executes 20 steps before replanning.

## 4.2 Extending Robot Demonstration Coverage

Each task combines 50 fixed robot demonstrations with 0, 25, 50, 75, or 100 converted human demonstrations, randomly sampled independently at each budget from its 100-clip pool without nesting. Policy architecture, training budget, and evaluation protocol remain fixed. TABLE 1 shows that KODA provides its largest gains on conditions missing from robot demonstrations. With 100 converted demonstrations, SONIC Robot-OOD success rises from 3.75% to 48.75% (+45 percentage points), and HEFT rises from 5.00% to 51.25% (+46.25 points). Robot-ID success also improves, from 62.5% to 72.5% for SONIC and from 68.75% to 76.25% for HEFT. Robot-OOD macro-average trends upward over tested budgets for both controllers. Individual-task and Robot-ID results vary at intermediate data sizes, so we focus on overall gains at the full data budget. KODA thus supplies useful supervision for additional conditions covered by human demonstrations without adding robot demonstrations. Fig. 4 illustrates converted data and real-robot execution in the Robot-OOD setting for cup-to-drawer. See Appendix F for the other three tasks. The same converted human demonstrations yield large Robot-OOD gains under both SONIC and HEFT, demonstrating cross-controller reuse, as shown in TABLE 1. The following ablations isolate observation alignment and control command generation.

## 4.3 Observation Alignment

We hold the converted demonstrations, state–action supervision, policy architecture, training budget, and evaluation protocol fixed across four tasks. No alignment retains the original human images. Fixed offset uses a 0.10 m vertical camera shift with ±0.05 m training-time perturbations, adapting the fixed-offset strategy of (Shi et al., 2026) to chest-mounted capture. Fixed offset and Motion-based reprojection use only reprojection and hole inpainting; Motion-based reprojection derives the target camera poses from the canonical robot motion. Full alignment additionally removes human arms and renders and composites the target robot. See Appendix D.3 for details.

Full alignment reaches 48.75% Robot-OOD success versus 23.75% with the adapted EgoHumanoid fixed-offset baseline (Shi et al., 2026) and 18.75% without alignment (TABLE 2). Motion-based reprojection reaches 37.5%. Fixed offset, Motion-based reprojection, and Full alignment yield Robot-ID point estimates of 63.75%/66.25%/72.5%, respectively.

## 4.4 Controller-Consistent State–Action Supervision

The variant without SCO directly uses the generated canonical whole-body motion as the tracking controller's reference, omitting optimization during control command generation while retaining Whole-Body Motion Generation and Robot Onboard View Generation. Both variants use identical canonical trajectories, observations, demonstrations, policy architecture, training budget, and evaluation protocol. Since body states come from controller rollouts, the comparison evaluates state–action supervision as a whole. See Appendix D.3 for details.

**Table 2** Effect of observation alignment on SONIC policy success (%). The four variants use the same demonstrations, state–action supervision, policy architecture, training budget, and evaluation protocol. Robot-ID and Robot-OOD are reported separately for each variant. Bold marks the highest success rate for each task and condition, including ties.

| Task | No alignment | | Fixed offset | | Motion-based reprojection | | Full alignment | |
|---|---|---|---|---|---|---|---|---|
| | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD | Robot-ID | Robot-OOD |
| Cup-to-basket | 50 | 15 | 60 | 15 | 65 | 25 | **70** | **40** |
| Parcel-to-shelf | 85 | 35 | 80 | 45 | 85 | 65 | **90** | **75** |
| Basket-to-shelf | **65** | 20 | 60 | 25 | 60 | 35 | **65** | **45** |
| Cup-to-drawer | 50 | 5 | 55 | 10 | 55 | 25 | **65** | **35** |
| Macro-average | 62.5 | 18.75 | 63.75 | 23.75 | 66.25 | 37.5 | **72.5** | **48.75** |

**Table 3** Effect of controller-specific command optimization on SONIC policy success (%). Both variants use 50 robot and 100 converted-human demonstrations per task, with identical canonical trajectories and synthesized observations. Without SCO omits optimization only during Control Command Generation; With SCO uses optimized commands. Policy architecture, training budget, and evaluation protocol are held fixed. Bold marks the higher success rate for each task and condition.

| Task | Robot-ID | | Robot-OOD | |
|---|---|---|---|---|
| | Without SCO | With SCO | Without SCO | With SCO |
| Cup-to-basket | 25 | **70** | 0 | **40** |
| Parcel-to-shelf | 70 | **90** | 10 | **75** |
| Basket-to-shelf | 50 | **65** | 0 | **45** |
| Cup-to-drawer | 50 | **65** | 20 | **35** |
| Macro-average | 48.75 | **72.5** | 7.5 | **48.75** |

SCO: sampling-based command optimization. Macro-average weights the four tasks equally.

TABLE 3 shows that optimized control command generation raises macro-average success from 48.75% to 72.5% Robot-ID and from 7.5% to 48.75% Robot-OOD, gains of 23.75 and 41.25 percentage points. Both metrics improve on all four tasks. Thus, controller-command generation also affects real-robot Robot-ID performance. Without this optimization, Robot-ID success falls below the 62.5% robot-only baseline in TABLE 1; the complete pipeline reaches 72.5%. Controller-consistent supervision thus matters to whether the added data benefits the policy. Together, the ablations support aligning both observations and state–action supervision with deployment.

## 4.5 Motion Fidelity and Controller Trackability

Motion-generation entries in TABLE 4 average aggregate errors from LAFAN (Harvey et al., 2020) and 400 egocentric training clips. LAFAN provides a check on highly dynamic motions; both controllers share evaluation data.

Wrist errors decrease to 3.03 cm/0.1106 rad with SCO and 0.33 cm/0.0296 rad after Mink refinement. Controller-specific SCO improves wrist/camera tracking except SONIC camera orientation (+0.0024 rad). Replaying the motions before and after Mink refinement with the same SONIC tracker on both sources yields comparable errors: pelvis-relative non-arm/wrist positions 2.60/1.59 → 2.73/1.75 cm, torso orientation 0.0588 → 0.0579 rad, and non-arm joint RMSE 0.0750 → 0.0784 rad. Metric and replay details are in Appendix D.4. TABLES 2 and 3 show task benefits from observation alignment and controller-specific supervision. TABLE 1 supports the reusable records' training utility across both controllers despite residual tracking mismatch.

**Table 4** Tracking errors for whole-body motion generation and controller rollouts. Position and orientation are reported separately in centimeters and radians; lower is better. Generation errors use the source motion as the reference, while rollout errors use the fixed canonical motion. Bold marks the smallest reported value for each target and error type, including ties.

| Stage | Target | Direct | | SCO | | SCO+IK | |
|---|---|---|---|---|---|---|---|
| | | Position (cm) | Orientation (rad) | Position (cm) | Orientation (rad) | Position (cm) | Orientation (rad) |
| Whole-body motion generation | Wrist | 15.02 | 0.2046 | 3.03 | 0.1106 | **0.33** | **0.0296** |
| | Torso | 16.27 | 0.1302 | **5.39** | **0.0930** | **5.39** | **0.0930** |
| SONIC rollout | Wrist | 61.50 | 0.1777 | **2.88** | **0.1292** | – | – |
| | Camera | 60.90 | **0.0784** | **3.20** | 0.0808 | – | – |
| HEFT rollout | Wrist | 19.61 | 0.0677 | **2.94** | **0.0552** | – | – |
| | Camera | 19.51 | 0.0426 | **2.84** | **0.0337** | – | – |

Direct uses unoptimized references. SCO uses CEM-optimized commands; SCO+IK adds Mink arm refinement during whole-body motion generation. A dash (–) denotes an unreported metric or variant.

## 5 Limitations

KODA does not yet fully reproduce fine-grained hand–object interactions, limiting their fidelity and physical feasibility. In the future, we plan to incorporate real-to-sim scene reconstruction and contact modeling to better recover object geometry and interaction dynamics. We will also broaden human data sources beyond egocentric recordings, including third-person videos and motion capture, converting diverse human demonstrations into canonical whole-body records for controller-specific state–action generation and policy training.

## 6 Conclusion

KODA converts egocentric human demonstrations into canonical whole-body motion and onboard observations, then generates controller-specific state–action supervision for VLA training. Reusing these records with fixed robot data improves VLA task success under SONIC and HEFT on four G1 tasks, especially on conditions covered only by human demonstrations. Reduced tracking errors and co-training gains despite residual mismatch support expanding robot-policy coverage.

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# Appendix

Appendix A expands contact-aware retargeting, sparse command optimization, and arm refinement. Appendix B specifies coordinate and visibility conventions. Appendix C details controller-specific search, action encoding, and state construction. Appendix D clarifies experimental conditions, comparisons, and metric references. Appendix E presents complete-clip demonstration timelines. Appendix F provides Robot-ID, KODA, and Robot-OOD comparisons for the remaining three tasks.

## A Whole-Body Motion Generation Details

This appendix expands Sec. 3.2. Source-camera and bilateral hand references share a metric world frame. Retargeted wrist and finger tracks $w^{\mathrm{ret}}_{1:T}, f^{\mathrm{ret}}_{1:T}$ supply the hand references; $r^{\mathrm{ref}}_{1:T}$ denotes the desired torso and wrist poses. SCO produces realized whole-body motion $q^{\mathcal{R},\mathrm{SCO}}_{1:T}$. Arm refinement yields the canonical floating-base, body, and hand sequence $q^{\mathcal{R}}_{1:T}$ while retaining the hand references and non-arm motion.

### A.1 Contact-aware hand retargeting

*Initialization and contact evidence.* HAWOR (Zhang et al., 2025) supplies world-space MANO hand estimates. Retargeting following AnyTeleop (Qin et al., 2023) uses palm geometry and a calibrated wrist-to-palm transform to initialize each robot wrist and its active finger joints. Distal mimic joints are determined by the coupling map, so they are not independent optimization variables. A correspondence between selected human landmarks and robot-hand points provides the geometric basis for refinement. The correspondences express analogous locations on the two hands; they do not require their shapes or articulation ranges to coincide.

The hand/forearm segmentation removes human-hand pixels from the depth samples used as object evidence. Nearby valid non-hand depths provide local object surface points. Multi-finger proximity, depth validity, and local surface consistency identify candidate contact frames, which are temporally filtered. These checks determine where geometric refinement is supported by the observation. When an entire sequence has no reliable contact, the initialized hand trajectory is retained. Contact evidence is used to recover spatial hand–object relationships; it is not a measurement of contact force.

*Local wrist and finger corrections.* At one contact frame, let $(p^0, R^0)$ be the initialized wrist pose and $f^0$ the active finger-joint vector for one hand. In the implemented six-active-joint hand model, $\delta p, \delta\omega \in \mathbb{R}^3$ and $\delta f \in \mathbb{R}^6$. We optimize $\xi = (\delta p, \delta\omega, \delta f) \in \mathbb{R}^{12}$ using

$$\begin{aligned} p(\xi) &= p^0 + R^0 \delta p, \\ R(\xi) &= R^0 \,\mathrm{Exp}([\delta\omega]_\times), \\ f(\xi) &= \mathrm{clip}(f^0 + \delta f, f_{\min}, f_{\max}). \end{aligned} \tag{A.1}$$

The initialized position $p^0$ is in the world frame, and $R^0$ maps wrist coordinates to world coordinates. Both wrist corrections are expressed relative to the initialized wrist frame. Translations are in meters; rotation vectors and finger-joint angles are in radians. The clip is componentwise, and $f_{\min}, f_{\max} \in \mathbb{R}^6$ are the active-joint limits in the robot's MJCF model. Here $[\delta\omega]_\times$ is the skew-symmetric matrix associated with the rotation vector, and Exp maps it to a valid rotation. Forward kinematics evaluates the robot landmarks under the corrected wrist and finger configuration, including the fixed mimic-joint relationships. Thus wrist placement and finger articulation affect the same geometric objective.

*Shared interaction geometry.* Let $P^h$ be the selected human landmarks, $P^{\mathcal{R}}(\xi)$ their robot counterparts, and $O$ the object samples for this frame. Following the interaction-mesh construction of OmniRetarget (Yang et al., 2025a), the ordered vertex arrays $V^h = [P^h; O]$ and $V^{\mathcal{R}}(\xi) = [P^{\mathcal{R}}(\xi); O]$ share vertex correspondences and connectivity. The implementation constructs adjacency from a Delaunay tetrahedralization of the human landmarks and object samples, with a nearest- neighbor fallback when tetrahedralization fails. Connectivity is then held fixed throughout the frame's optimization and reused for the robot vertices.

Let $v_i \in \mathbb{R}^3$ be the world-frame position of the $i$th vertex of $V$. For each vertex with a nonempty neighbor set $\mathcal{N}(i)$, its descriptor is

$$L_i(V) = v_i - \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} v_j. \tag{A.2}$$

The uniform weights sum to one. Consequently, adding a common translation to the vertices leaves their descriptors unchanged. Hand–object relationships still constrain the wrist because the observed object vertices remain fixed while the robot-hand vertices move. Object descriptors can also change when their neighbors include hand vertices, even though the object positions themselves are constant. The descriptor discrepancy therefore includes both hand and object rows:

$$\xi^* = \arg\min_{\xi \in \mathcal{A}} \lambda_L \sum_{i \in \mathcal{V}} \left\| L_i(V^{\mathcal{R}}(\xi)) - L_i(V^h) \right\|_2^2 + \mathcal{L}_{\text{reg}}(\xi). \tag{A.3}$$

Here $\mathcal{V}$ indexes corresponding vertices and $\lambda_L$ weights descriptor matching. Let $\xi^- = (\delta p^-, \delta\omega^-, \delta f^-)$ denote the preceding solved contact-frame correction, with $\xi^- = 0$ for the first contact frame, and define $[a]_+ = \max(a, 0)$. The regularizer is

$$\begin{aligned}
\mathcal{L}_{\text{reg}}(\xi) = {} & \alpha_w^2 \left( \|\delta p\|_2^2 + \|\delta\omega\|_2^2 \right) + \alpha_f^2 \|\delta f\|_2^2 \\
& + \beta_w^2 \left( \|\delta p - \delta p^-\|_2^2 + \|\delta\omega - \delta\omega^-\|_2^2 \right) \\
& + \beta_f^2 \|\delta f - \delta f^-\|_2^2 \\
& + \gamma^2 \left( [\|\delta p\|_2 - b_t]_+^2 + [\|\delta\omega\|_2 - r_{\max}]_+^2 \right).
\end{aligned}$$

The coefficients $\alpha_w, \alpha_f$ weight wrist and finger anchoring; $\beta_w, \beta_f$ weight temporal changes in those corrections; and $\gamma$ weights the soft magnitude limits. These coefficients multiply residuals, so their squares appear in the objective. The first two terms anchor the correction to its initialization, the next two penalize changes from the preceding contact solution, and the last term softly limits translation and rotation magnitudes. We use the following default weights:

| Term | Parameter | Value |
|---|---|---|
| Laplacian descriptor loss | $\lambda_L$ | 10 |
| Wrist anchor residual | $\alpha_w$ | 0.10 |
| Finger anchor residual | $\alpha_f$ | 0.018 |
| Wrist temporal residual | $\beta_w$ | 12 |
| Finger temporal residual | $\beta_f$ | 2 |
| Soft-limit residual | $\gamma$ | 40 |

The correction box at frame $t$ is

$$\begin{aligned}
\mathcal{A} = \{ \xi : {} & |\delta p_i| \le b_t, \quad |\delta\omega_i| \le r_{\max} \quad (i = 1, 2, 3), \\
& |\delta f_j| \le \delta f_{\max} \quad (j = 1, \ldots, 6) \}.
\end{aligned}$$

Here $b_t$ is the frame's translation bound in meters, $r_{\max}$ is the rotation-vector coordinate bound in radians, and $\delta f_{\max}$ is the finger-correction coordinate bound in radians. The default angular bounds are $r_{\max} = \pi/15$ rad ($12^\circ$) and $\delta f_{\max} = 0.28$ rad. These are componentwise box bounds; the rotation-vector norm also enters the soft magnitude penalty above. Active-joint limits are applied separately by the clipping in Eq. (A.1). Let $d_{0,t}$ be the world-frame distance from the initialized palm center to the centroid of the object samples used in the Laplacian graph, with $d_{0,t} = 0$ for an empty sample set. The default translation bound is

$$b_t = \begin{cases} \min(d_{0,t}, 0.040 \text{ m}), & d_{0,t} \text{ is finite and } d_{0,t} \ge 0.025 \text{ m}, \\ 0.015 \text{ m}, & \text{otherwise.} \end{cases}$$

These coordinate bounds and soft magnitude penalties restrict displacement when local depth evidence is incomplete.

*Numerical refinement and temporal propagation.* The contact solver stacks the descriptor, anchor, temporal, and soft-limit residuals into a vector $\rho(\xi)$ so that $\|\rho(\xi)\|_2^2$ equals the objective in Eq. (A.3). In particular, descriptor differences are multiplied by $\sqrt{\lambda_L}$; the remaining residual blocks use $\alpha_w, \alpha_f, \beta_w, \beta_f, \gamma$ as defined above. Let $J_\rho$ denote the Jacobian of this weighted residual vector with respect to $\xi$. The fixed Laplacian matrix $L$ satisfies $(LV)_i = L_i(V)$, and $I_3$ is the $3 \times 3$ identity. The unweighted descriptor Jacobian follows by applying $L \otimes I_3$ to the robot-vertex Jacobian; object-vertex derivatives are zero. Translation derivatives follow the wrist transform analytically. Rotation and finger-joint derivatives use forward finite differences with an increment of $10^{-4}$ rad. Each iteration computes a damped linearized step,

$$\delta\xi = -(J_\rho^\top J_\rho + \lambda_d I)^{-1} J_\rho^\top \rho, \tag{A.4}$$

where $I$ is the $12 \times 12$ identity and $\lambda_d$ is the damping coefficient, set to $10^{-4}$. The solver first clips the step against the remaining coordinate bounds, then rescales it if its Euclidean norm exceeds the default trust-region radius of 0.2. This norm uses the correction coordinates in meters and radians without additional coordinate normalization. The corrected pose is re-evaluated before the next iteration. The default iteration limits are 20 for the first contact frame and 10 for subsequent contact frames. Iterations stop earlier when successive squared-residual costs satisfy NumPy's default `isclose` test or when the clipped and rescaled step has norm below $2 \times 10^{-4}$; in the latter case, the small step is applied before termination. The preceding contact solution initializes the next contact-frame solve.

Within the contact optimizer, correction candidates are propagated to the target clock by linear interpolation of translation and finger offsets and shortest-path interpolation of the correction rotations on $SO(3)$ between contact intervals. Before the first contact, the correction ramps from zero at the sequence start to the first contact correction; after the last contact it ramps to zero at the sequence end. These ramps apply when the corresponding non-contact frames exist. With no contact frames, all corrections are zero. The propagated correction coordinates are then smoothed with a separate Savitzky–Golay filter whose default window contains nine frames and whose polynomial degree is two, using polynomial edge evaluation (`interp` mode). Sequences shorter than nine frames are left unchanged by this filter. Filtering does not impose the prefilter endpoint values as equality constraints. This correction filter is distinct from the sparse-reference filter described in Sec. A.3. The resulting $f_{1:T}^{\mathrm{ret}}$ is retained as a continuous geometric hand reference; the grasp and release command construction is a separate operation described in Sec. 3.4.

For the tracking pipeline, the wrist reference is assembled from the candidate wrist poses over the interval between the first and last contact frames. Outside this interval, the first-contact residual is held before contact, and the last-contact residual is held after contact. The held quantities are the world-frame translation offset and the local rotation offset composed on the right of the nominal wrist orientation. They are applied relative to the nominal wrist pose at each frame, so holding these residuals does not fix the absolute wrist pose. This boundary rule applies to the injected wrist reference and does not replace the interpolated and smoothed finger trajectory. The assembled three-point reference is subsequently filtered as described in Sec. A.3.

## A.2 Sparse pose-command optimization

*Search space and decoding.* The source-camera poses are mapped to torso poses using the target robot's URDF camera-to-torso transform. Together with the retargeted wrists, these produce three ordered reference tracks: torso, left wrist, and right wrist. We extend ScaleBFM (Zeng et al., 2026) with a three-point tracking mode conditioned on the torso and bilateral wrist poses, and fine-tune the pretrained controller for this mode using ScaleBFM's training data. The resulting tracker receives these sparse pose commands and produces whole-body joint targets for PD control. SCO changes the sparse commands and evaluates their realized motion in simulation; it does not directly search all robot joint configurations.

At each replanning instant, $K = 10$ knots span a horizon of $H = 40$ control steps. For each tracked body $b$ and knot $k$, the latent contains translation and rotation-vector residuals $(z_{kb}^p, z_{kb}^R) \in \mathbb{R}^6$. The three bodies therefore contribute 18 coordinates per knot and 180 coordinates in total. Normalized coordinates are clipped to $[-1, 1]$ before physical scaling:

$$\delta p_{kb} = 0.055\ \mathrm{clip}(z_{kb}^p, -1, 1), \qquad \delta\omega_{kb} = 0.0825\ \mathrm{clip}(z_{kb}^R, -1, 1). \tag{A.5}$$

**Table A.1** SCO settings for Whole-Body Motion Generation. The residual and proposal scales apply to this stage's three-point pose-command space.

| Quantity | Value |
|---|---|
| Control frequency | 50 Hz |
| Planning horizon $H$ / executed prefix $\Delta$ | 40 / 10 steps |
| Temporal knots $K$ / latent dimension $d$ | 10 / 180 |
| Candidates $N$ / CEM updates $I$ / elites $E$ | 80 / 8 / 10 |
| Initial standard deviation / floor | 1.6 / 0.075 |
| Translation / rotation scale | 0.055 m / 0.0825 rad |
| Position / orientation cost weight | 1.0 / 0.1 |
| Torso / left wrist / right wrist weight | 0.25 / 1.0 / 1.0 |

Translation is in meters and rotation is in radians. Both residuals are expressed in the world frame. Let $(p^{\mathrm{ref}}_{\tau b}, R^{\mathrm{ref}}_{\tau b})$ denote the unmodified reference pose of tracked body $b$ at horizon step $\tau \in \{1, \dots, H\}$, where $p^{\mathrm{ref}}_{\tau b}$ is its world-frame position and $R^{\mathrm{ref}}_{\tau b}$ maps body-frame vectors to the world frame.

The knots are uniformly spaced at $\tau_k = 1 + (k-1)(H-1)/(K-1)$ for $k = 1, \dots, K$. Let $A_{\tau k}$ be the piecewise-linear interpolation weight from knot $k$ to horizon step $\tau$. For $\tau_j \le \tau \le \tau_{j+1}$, the only potentially nonzero weights are

$$A_{\tau j} = \frac{\tau_{j+1} - \tau}{\tau_{j+1} - \tau_j}, \qquad A_{\tau,j+1} = \frac{\tau - \tau_j}{\tau_{j+1} - \tau_j}.$$

The decoder interpolates the knot residuals and constructs the sparse pose commands as

$$\begin{aligned} \delta p_{\tau b} &= \sum_{k=1}^{K} A_{\tau k}\, \delta p_{kb}, \quad & \delta \omega_{\tau b} &= \sum_{k=1}^{K} A_{\tau k}\, \delta \omega_{kb}, \\ p^{\mathrm{cmd}}_{\tau b} &= p^{\mathrm{ref}}_{\tau b} + \delta p_{\tau b}, \quad & R^{\mathrm{cmd}}_{\tau b} &= \mathrm{Exp}([\delta \omega_{\tau b}]_\times)\, R^{\mathrm{ref}}_{\tau b}. \end{aligned} \tag{A.6}$$

Here $[\cdot]_\times$ is the skew-symmetric matrix operator satisfying $[v]_\times x = v \times x$, and Exp is the matrix exponential. The rotation-vector residuals are interpolated before conversion to rotation matrices; left multiplication applies the resulting rotation correction in the world frame.

The decoded poses $(p^{\mathrm{cmd}}_{\tau b}, R^{\mathrm{cmd}}_{\tau b})$ are supplied to the three-point tracker. Their realized link poses are denoted by $(\widehat{p}_{\tau b}, \widehat{R}_{\tau b})$ in the tracking objective below and are evaluated against the unchanged reference poses. Joint references, pelvis guidance, and other body references are not independently perturbed by this search. Their realized motion is determined by the tracker and simulator.

*Tracking objective.* All candidates start from the same current simulator state and are scored against the unmodified reference $\mathbf{r}^{\mathrm{ref}}$. Let $(\widehat{p}_{\tau b}, \widehat{R}_{\tau b})$ denote a candidate's realized link pose at horizon step $\tau$, and let $(p^{\mathrm{ref}}_{\tau b}, R^{\mathrm{ref}}_{\tau b})$ be the corresponding desired pose. The objective is

$$\begin{aligned} J &= \sum_{\tau=1}^{H} \omega_\tau \sum_{b \in \mathcal{B}} \eta_b \Big[ \|\widehat{p}_{\tau b} - p^{\mathrm{ref}}_{\tau b}\|_2^2 + 0.1\ \mathrm{angle}\Big(\widehat{R}_{\tau b}^{\top} R^{\mathrm{ref}}_{\tau b}\Big)^2 \Big], \\ \omega_\tau &= \frac{H - \tau + 1}{\sum_{j=1}^{H} (H - j + 1)}. \end{aligned} \tag{A.7}$$

The body set $\mathcal{B}$ contains the torso and bilateral wrists. angle gives the relative rotation angle in radians. Link weights are $\eta_{\mathrm{torso}} = 0.25$ and $\eta_{\text{left wrist}} = \eta_{\text{right wrist}} = 1$, without normalization across links. The pelvis has zero weight; joint, velocity, reference-deformation, and terminal penalties are inactive for this stage. The decreasing time weights prioritize the prefix that will be executed next. Scoring the realized motion against the unchanged target ensures that changing the input command alone cannot be counted as improved tracking.

*Diagonal CEM and simulator failures.* CEM (Pinneri et al., 2021) maintains a mean $\mu$ and a coordinate-wise standard deviation $\sigma$, initialized to zero and 1.6, respectively. Each iteration samples $N$ latent vectors,

$$z^{(n)} = \mu + \sigma \odot \varepsilon^{(n)}, \qquad \varepsilon^{(n)} \sim \mathcal{N}(0, I_d), \tag{A.8}$$

then decodes and evaluates the corresponding commands in parallel rollouts. For the $E$ lowest-cost candidates indexed by $\mathcal{E}$, the proposal update is

$$\begin{aligned} \mu_i^+ &= \frac{1}{E} \sum_{n \in \mathcal{E}} z_i^{(n)}, \\ \sigma_i^+ &= \max\left\{ 0.075, \sqrt{\frac{1}{E} \sum_{n \in \mathcal{E}} \left( z_i^{(n)} - \mu_i^+ \right)^2} \right\}. \end{aligned} \tag{A.9}$$

The divisor is $E$, giving the population standard deviation of the elite coordinates. The proposal is diagonal, so the update does not estimate a full covariance from a small elite set. Sampling operates in normalized latent coordinates; physical clipping is applied by the decoder.

Non-finite simulator measurements or objective values are handled by assigning a large finite failure cost before candidate ranking. CEM selects the $E$ lowest-cost samples from all $N$ candidates without a separate candidate-validity filter. The same failure-cost convention is used when comparing the nominal command, the best sampled candidate, and the final proposal mean. Tracking-error-based rejection is applied to complete episodes after SCO, as described below.

*Receding-horizon execution.* One replanning update proceeds as follows:

1. Construct the next reference horizon and initialize the diagonal proposal. All candidates share the current simulated starting state.
2. Sample, decode, and roll out $N$ sparse-command candidates. Score them against the fixed target using the failure-cost convention above, and update the proposal using the $E$ lowest-cost samples. Repeat for $I$ updates.
3. Compare the nominal command, the best sampled candidate, and the final proposal mean using their rollout costs, and retain the lowest-cost choice.
4. Execute its first $\Delta = 10$ steps, record the realized configurations, and continue planning from the resulting simulator state.

At 50 Hz, the horizon covers 0.80 s and each retained prefix covers 0.20 s. The candidate count $N$ denotes sampled proposals per iteration; nominal and final-mean evaluations are distinct comparison candidates. Concatenating executed prefixes yields $q_{1:T}^{\mathcal{R},\mathrm{SCO}}$. Advancing from the realized state carries tracking deviations into the next optimization instead of restarting each horizon at an ideal reference pose.

*Episode acceptance before arm refinement.* After SCO has generated the complete trajectory $q_{1:T}^{\mathcal{R},\mathrm{SCO}}$, we evaluate its mean wrist-position tracking error before applying Mink arm refinement. Let $\mathcal{B}_{\mathrm{wrist}}$ contain the left and right wrist links. For each wrist $b$, let $p_{tb}^{\mathrm{SCO}}$ be its world-frame position obtained by forward kinematics from $q_t^{\mathcal{R},\mathrm{SCO}}$, and let $p_{tb}^{\mathrm{ref}}$ be the corresponding unchanged wrist-position target used during SCO. The episode-level error is

$$\bar{e}_{\mathrm{wrist}} = \frac{1}{2T} \sum_{t=1}^{T} \sum_{b \in \mathcal{B}_{\mathrm{wrist}}} \left\| p_{tb}^{\mathrm{SCO}} - p_{tb}^{\mathrm{ref}} \right\|_2 . \tag{A.10}$$

This is the arithmetic mean of the Euclidean position errors over both wrists and all frames of the complete episode. Episodes with $\bar{e}_{\mathrm{wrist}} > 0.075$ m are discarded; retained episodes proceed to Mink arm refinement. This acceptance criterion uses wrist-position error only and is separate from the receding-horizon optimization objective in Eq. A.7.

### A.3 Temporal smoothing and constrained arm refinement

*Filtering on the rotation manifold.* Savitzky–Golay filtering (Schafer, 2011) smooths the three-point reference positions in the common Z-up world frame, using a five-frame window and a quadratic polynomial under the default settings. Orientation filtering uses local tangent coordinates so that smoothing does not produce invalid rotation matrices. For a window around frame $t$, the neighboring rotations $R_j$ are represented relative to $R_t$ as $\mathrm{Log}(R_t^\top R_j)^\vee$, using the principal rotation vector with norm at most $\pi$. A local polynomial fit using the actual timestamps is evaluated at the timestamp of frame $t$ to obtain $\delta\omega_t^{\mathrm{SG}}$, and the filtered orientation is

$$R_t^{\mathrm{SG}} = R_t \,\mathrm{Exp}\big([\delta\omega_t^{\mathrm{SG}}]_\times\big). \tag{A.11}$$

The superscript $\vee$ converts a skew-symmetric matrix to its rotation vector. Near a segment boundary, the full window shifts inside the segment and the fitted polynomial is still evaluated at the target frame's timestamp. The implementation splits the sequence where an adjacent timestamp gap exceeds three times the median sampling interval of the full sequence; segments shorter than five frames are left unchanged. Position filtering uses the same actual-time polynomial fit and boundary handling. Position and orientation filtering therefore use the same temporal reference while respecting their different geometry. Kimodo (Rempe et al., 2026) provides an initial whole-body configuration from the first sparse reference frame; the checked configuration initializes whole-body simulation.

*Arm-only refinement and final assembly.* SCO reduces sparse-reference error through the controller's dynamics, but its realized wrists can still deviate from the retargeted manipulation reference. Temporal Mink IK (Zakka, 2024) addresses this residual by adjusting the arm chain, including the wrists. The optimized floating base, torso, and lower-body trajectory are held fixed, and the contact-aware finger trajectory is retained. Joint limits, per-step displacement and velocity bounds, and self-collision constraints restrict the arm updates. A posture penalty discourages unnecessary departures from the SCO configuration. Temporal restrictions are applied across successive refined states so that reducing a frame's wrist error does not permit an unconstrained joint jump.

The final configuration combines the refined arms, retargeted fingers, and fixed non-arm motion. It supplies both the canonical motion reference and the body trajectory used for robot-view rendering. Holding the non-arm configuration fixed preserves those kinematic tracks; changes to the arms can still affect the response of a whole-body controller. The replay comparison in Sec. 4.5 therefore evaluates trackability before and after refinement. These measurements assess the demonstrated trajectories under the evaluated controller, without assuming that inverse kinematics alone establishes dynamic feasibility for arbitrary motions or deployment interfaces.

## B Coordinate Conventions and Observation Synthesis

This appendix expands the geometric operations in Sec. 3.3. The recovered human scene and the rendered robot must be expressed in the same target-camera frame before their depths can be compared. The target camera is obtained from the canonical whole-body motion, so scene reprojection and robot rendering use one pose at each time index.

### B.1 Reference frames and the canonical record

Let $T_{A\leftarrow B} \in \mathrm{SE}(3)$ map homogeneous coordinates in frame $B$ into frame $A$. The world frame is denoted by $W$, the robot torso frame by $B$, and its camera frame by $C$. A camera-to-world pose has the form

$$C_t = \begin{bmatrix} R_t & p_t \\ 0^\top & 1 \end{bmatrix}, \qquad C_t^{-1} = \begin{bmatrix} R_t^\top & -R_t^\top p_t \\ 0^\top & 1 \end{bmatrix}. \tag{B.1}$$

Here $R_t \in \mathrm{SO}(3)$ maps camera-frame vectors to the world frame, and $p_t \in \mathbb{R}^3$ is the camera origin expressed in world coordinates. Thus $C_t$ maps camera-frame points to world coordinates, whereas $C_t^{-1}$ maps world points into the camera frame for projection. All depths and translations in the geometric stages use a consistent metric scale. Image coordinates and calibrated intrinsics must refer to the same image size. The control clock

provides the common time index for source observations, recovered geometry, canonical configurations, and the synthesized observations stored with them.

Write $T_{B\leftarrow C}$ for the fixed camera-to-torso transform from the target robot's calibration and URDF. We express the source and robot camera poses in a common world frame and use a consistent optical-frame convention. To construct the torso reference, we take the source camera pose $C_t^h$ as the desired robot camera pose. The torso reference and the camera pose derived from canonical robot motion are then given by

$$\begin{aligned} T_{W\leftarrow B}^{\mathrm{ref}}(t) &= C_t^h T_{B\leftarrow C}^{-1}, \\ C_t^{\mathcal{R}} &= T_{W\leftarrow B}(q_t^{\mathcal{R}}) T_{B\leftarrow C}. \end{aligned} \tag{B.2}$$

Here $T_{W\leftarrow B}(q_t^{\mathcal{R}})$ is the torso-to-world transform obtained by forward kinematics at the canonical robot configuration. The first line expresses the torso target $p_t^{\mathrm{torso}}$ from Sec. 3.2 as a homogeneous transform; the second uses the generated configuration to define the rendering camera. The resulting robot camera trajectory can differ from the source camera trajectory because whole-body tracking may not exactly realize the torso reference under the robot's kinematic and control constraints. Arm refinement changes the rendered arm geometry while leaving the torso and camera trajectory fixed. In particular, the observation stage uses the second line rather than substituting the source camera path or a constant vertical displacement.

The canonical record retains $(I_t^{\mathcal{R}}, q_t^{\mathcal{R}}, \ell)$ for a fixed target embodiment and camera convention. Its independence from the deployment interface means that these quantities are retained when the subsequent state–action mapping changes between SONIC and HEFT. A tracking controller is still used to construct the canonical motion. The body state recorded during deployment-controller replay can differ from the canonical configuration; the interface appendix specifies this distinction in detail.

## B.2 Metric back-projection and target-camera visibility

Let $K_h$ and $K_{\mathcal{R}}$ denote the source and target intrinsics. The following equations assume distortion-free pinhole image coordinates and zero-skew intrinsics, with source RGB and depth registered to the same pixel grid. A source pixel is written as $u = (u_x, u_y)$, with homogeneous coordinate $\bar{u} = (u_x, u_y, 1)^\top$. EfficientSAM3 (Zeng et al., 2025) provides the source arm/hand mask. Invalidating the masked depth before projection prevents the source limbs from entering the reconstructed scene. Only unmasked source pixels with finite, positive optical-axis depth are back-projected. For each such sample,

$$\begin{aligned} p_h(u) &= \widetilde{D}_t^h(u) K_h^{-1} \bar{u}, \\ \begin{bmatrix} p_{\mathcal{R}}(u) \\ 1 \end{bmatrix} &= (C_t^{\mathcal{R}})^{-1} C_t^h \begin{bmatrix} p_h(u) \\ 1 \end{bmatrix}, \\ v_x(u) &= f_x^{\mathcal{R}} \frac{[p_{\mathcal{R}}(u)]_x}{[p_{\mathcal{R}}(u)]_z} + c_x^{\mathcal{R}}, \qquad v_y(u) = f_y^{\mathcal{R}} \frac{[p_{\mathcal{R}}(u)]_y}{[p_{\mathcal{R}}(u)]_z} + c_y^{\mathcal{R}}. \end{aligned} \tag{B.3}$$

Here $\widetilde{D}_t^h$ is the source depth after invalidating the human arm/hand mask. The points $p_h(u)$ and $p_{\mathcal{R}}(u)$ are expressed in the source-camera and target-camera frames, respectively; $v(u) = (v_x(u), v_y(u))$ is the projected target-image location. The quantities $f_x, f_y$ are focal lengths in pixels and $c_x, c_y$ are principal-point coordinates. The source depth is a $z$ coordinate, rather than the Euclidean distance from the camera center. After transformation, the target depth used for visibility is $[p_{\mathcal{R}}(u)]_z$, rather than the original source depth. Valid projected samples have finite coordinates and positive target $z$ and contribute only inside the target image.

Forward splatting maps source colors into the target grid. Under the default settings, each projected location is rounded to the nearest integer pixel center and splatted over its $3 \times 3$ neighborhood, clipped to the target image. Let $\mathcal{S}_t(v)$ denote the valid source samples whose projected splats cover target pixel $v$. The metric z-buffer selects the smallest target-camera depth among the valid reprojected samples covering each target pixel,

$$D_t^{\mathrm{geo}}(v) = \min_{u \in \mathcal{S}_t(v)} [p_{\mathcal{R}}(u)]_z, \qquad V_t^{\mathrm{geo}}(v) = \mathbf{1}[\mathcal{S}_t(v) \neq \emptyset]. \tag{B.4}$$

Thus, $V_t^{\mathrm{geo}}$ indicates valid reprojected scene support; visibility is resolved among the available source samples. The minimum is used only where $V_t^{\mathrm{geo}} = 1$; elsewhere the depth is invalid. Color is taken from a sample

at the nearest surface, with numerical ties resolved consistently. This visibility ordering prevents a farther surface from overwriting a nearer one merely because it is processed later. The geometric hole mask is $H_t^{\text{geo}} = 1 - V_t^{\text{geo}}$. It includes areas exposed by source-arm removal and areas that were not observed from the original viewpoint. Reprojection changes where visible source evidence is placed; it does not supply the appearance of these newly exposed regions.

### B.3 Robot rendering, compositing, and restricted completion

MuJoCo renders robot color and depth from $q_t^{\mathcal{R}}$ with the same $C_t^{\mathcal{R}}$ used in Eq. B.3. The rendered depth is represented in the same metric target-camera convention before visibility comparison. Write $V_t^{\text{rob}}$ for its binary validity mask. The visible-robot mask is defined at every target pixel by

$$M_t^{\text{rob}}(v) = \begin{cases} \mathbf{1}[D_t^{\text{rob}}(v) < D_t^{\text{geo}}(v)], & (V_t^{\text{rob}}(v), V_t^{\text{geo}}(v)) = (1,1), \\ 1, & (V_t^{\text{rob}}(v), V_t^{\text{geo}}(v)) = (1,0), \\ 0, & (V_t^{\text{rob}}(v), V_t^{\text{geo}}(v)) = (0,1), \\ 0, & (V_t^{\text{rob}}(v), V_t^{\text{geo}}(v)) = (0,0). \end{cases} \tag{B.5}$$

If only one surface is valid, that surface supplies the pixel. When neither is available, the pixel remains a hole. Neighboring valid scene depths may supply local depth context inside geometric holes, as described in the main pipeline; they do not replace the depth comparison at pixels with measured scene support. The visible-robot mask, rather than the rendered silhouette alone, controls the composite:

$$I_t^{\text{comp}} = M_t^{\text{rob}} I_t^{\text{rob}} + (1 - M_t^{\text{rob}}) I_t^{\text{geo}}, \qquad H_t^{\text{rem}} = H_t^{\text{geo}}(1 - M_t^{\text{rob}}). \tag{B.6}$$

Products here are pixel-wise, with binary masks broadcast over color channels. The color arrays $I_t^{\text{rob}}$ and $I_t^{\text{geo}}$ use finite zero placeholders wherever their respective validity masks are zero, so masked arithmetic is defined even at holes. A branch-based selection of the valid color gives the same composite. In particular, $H_t^{\text{rem}} = (1 - V_t^{\text{geo}})(1 - V_t^{\text{rob}})$: neither source supplies a pixel precisely when both validity masks are zero. A foreground scene object can occlude the robot, and a nearer robot surface can occlude the reprojected scene. The residual-hole formula is equivalent to the set difference in the main text when each binary mask is identified with its support: only pixels uncovered by both sources remain to be completed.

DiffuEraser (Li et al., 2025) supplies appearance in $H_t^{\text{rem}}$. Denoting its completed prediction by $I_t^{\text{fill}}$, the final observation can be expressed as

$$I_t^{\mathcal{R}} = (1 - H_t^{\text{rem}}) I_t^{\text{comp}} + H_t^{\text{rem}} I_t^{\text{fill}}. \tag{B.7}$$

Thus, valid scene and rendered-robot pixels remain fixed. Completion has a restricted appearance role after geometric visibility has been established; it does not change the canonical configuration, target camera pose, or action labels. The completed observation is stored with the same $q_t^{\mathcal{R}}$ and instruction to form the canonical episode.

For the reported 400 egocentric training clips, source depth is recorded by the iPhone. The RGB-only route in the framework instead uses metric depth from Depth Anything 3 (Lin et al., 2025) before the same geometric operations. These are alternative sources for $D_t^h$, rather than two separately evaluated training-data conditions. The observation ablation in Sec. 4.3 compares how views are aligned while keeping the converted motion and state–action supervision fixed.

## C Controller-Specific Command and State Interfaces

This appendix expands Sec. 3.4. For each deployment controller, KODA optimizes a body reference against the fixed canonical trajectory, evaluates the realized motion, and records state–action supervision. Table C.1 distinguishes the search representations from the quantities exported for VLA training.

### C.1 Shared Body-Reference Search

Let $m_t^{\text{can}}$ contain the canonical pelvis pose, the 29-dimensional body-joint vector $b_t^{\text{can}}$, and the associated link poses. The reference is represented on the 50 Hz control clock; joint velocities and link quantities are available

**Table C.1** Representations used in control-command generation. Dimensions refer to one knot or one time-indexed record, as indicated. The search coordinates are shared, while body-action labels depend on the controller.

| Quantity | Dimension | Contents and role |
|---|---|---|
| Search residual | 35 per knot | Pelvis translation and rotation-vector corrections, plus 29 body-joint corrections. |
| Horizon search | 350 | Ten residual knots interpolated over the horizon. |
| SONIC body label | 64 | Motion token from the optimized motion reference. |
| HEFT body label | $38 = 29 + 9$ | Absolute body-joint targets and pelvis code. |
| Hand action | 12 | Bilateral BrainCo targets, with grasp and release handled separately. |
| Complete action | 76 / 50 | SONIC / HEFT body label with 12 hand targets. |
| Policy state | $44 = 29 + 3 + 12$ | Rollout body joints, base-frame gravity, and retargeted hand angles. |

with the motion record. We use $\widetilde{m}_t$ for a candidate command reference and $\widehat{m}_t^{\kappa}$ for the motion realized by controller $\kappa$. These three quantities serve different roles: the canonical motion specifies the desired behavior, the candidate is supplied through the controller's reference interface, and the rollout determines its tracking cost.

For knot index $k = 1, \ldots, K$, each temporal knot contains normalized residuals

$$z_k = \left(z_k^p, z_k^\omega, z_k^b\right) \in \mathbb{R}^{3+3+29}, \qquad z = (z_1, \ldots, z_K), \quad K = 10. \tag{C.1}$$

Here $z_k^p, z_k^\omega \in \mathbb{R}^3$ and $z_k^b \in \mathbb{R}^{29}$ are dimensionless pelvis-translation, rotation-vector, and body-joint corrections, respectively. The resulting search vector $z \in \mathbb{R}^{350}$ is the variable sampled by CEM for either controller. At each iteration, CEM draws $N$ candidate vectors $z^{(n)}$, $n = 1, \ldots, N$, from a diagonal Gaussian distribution in this normalized knot space. Each sample is deterministically decoded into physical corrections $\delta_\tau^{(n)}$ at every control step through clipping, scaling, and temporal interpolation in Eq. C.2. Hands are excluded from this body-reference search and retain the retargeted geometric trajectory used for the hand-state and hand-command construction below. Let $S_\kappa \in \mathbb{R}^{35\times 35}$ denote the diagonal conversion from normalized coordinates to physical translation, rotation, and joint corrections. Both controllers use the same scales:

$$S_{\text{SONIC}} = S_{\text{HEFT}} = \operatorname{diag}(0.055 I_2,\ 0.0275,\ 0.0825 I_3,\ 0.1375 I_{29}),$$

where $I_d$ is the $d$-dimensional identity matrix. The first three scale entries are in meters; the rotation and joint entries are in radians. For control-step index $\tau$ within the current horizon, let $\delta_\tau = (\delta p_\tau, \delta\omega_\tau, \delta b_\tau) \in \mathbb{R}^{35}$ collect the dense physical corrections. With interpolation weights $A_{\tau k}$, they are

$$\delta_\tau = \sum_{k=1}^{K} A_{\tau k} S_\kappa \operatorname{clip}(z_k, -1, 1), \qquad \sum_k A_{\tau k} = 1. \tag{C.2}$$

The knots are uniformly spaced over the horizon, and the nonnegative weights $A_{\tau k}$ linearly interpolate neighboring knots. The clipping operation acts elementwise. Let $p_\tau^{\text{can}}$ and $R_\tau^{\text{can}}$ be the canonical pelvis position and pelvis-to-world rotation. Translation and rotation-vector corrections are expressed in the world frame and applied as

$$\begin{aligned}
\widetilde{p}_\tau &= p_\tau^{\text{can}} + \delta p_\tau, \\
\widetilde{R}_\tau &= \operatorname{Exp}([\delta\omega_\tau]_\times) R_\tau^{\text{can}}, \\
\widetilde{b}_\tau &= b_\tau^{\text{can}} + \delta b_\tau,
\end{aligned}$$

where $[v]_\times w = v \times w$ and Exp maps a skew-symmetric matrix to a rotation. Body-joint corrections are followed by elementwise clipping to joint limits when those limits are configured. Forward kinematics then recomputes the link poses from the perturbed pelvis and body joints, maintaining a single connected robot configuration.

The controller adapter encodes each candidate for simulation using SONIC motion encoding or the HEFT representation below. CEM searches the common physical reference space and evaluates each candidate through the selected controller. Its 350 search coordinates differ from the 64-dimensional SONIC token and 38-dimensional HEFT body label.

## C.2 Tracking Cost and Receding-Horizon Settings

The shared objective evaluates 29 body joints and 14 links: the pelvis, torso, and bilateral hip, knee, ankle, shoulder, elbow, and wrist links. For two motion states $m, m'$, let $b, b' \in \mathbb{R}^{29}$ be their body-joint angle vectors. Let $p_j, p'_j \in \mathbb{R}^3$ be the world-frame positions of link $j$, and $R_j, R'_j \in SO(3)$ its link-to-world rotations, for $j = 1, \ldots, 14$. Define

$$\begin{aligned} e_q(m, m') &= \frac{1}{29}\|b - b'\|_2^2, \\ e_p(m, m') &= \frac{1}{14}\sum_{j=1}^{14}\|p_j - p'_j\|_2^2, \\ e_R(m, m') &= \frac{1}{14}\sum_{j=1}^{14}\theta(R_j^\top R'_j)^2, \\ \theta(Q) &= \arccos\left(\mathrm{clip}\left(\frac{\mathrm{tr}(Q) - 1}{2}, -1, 1\right)\right). \end{aligned} \tag{C.3}$$

Here $Q \in SO(3)$ denotes a relative rotation and $\theta(Q) \in [0, \pi]$ its geodesic angle. Positions are in meters and angular quantities are in radians. Each link contributes equally to the link averages; joint error is averaged over the 29 joint coordinates. The orientation term is the squared geodesic angle of the relative rotation. Its quaternion implementation uses the absolute quaternion inner product, so the two quaternion signs representing the same rotation produce the same error.

For a horizon of $H$ steps, the per-scenario rollout cost is

$$\begin{aligned} J_\kappa = &\sum_{\tau=1}^{H}\Big[e_q(\widehat{m}_\tau^\kappa, m_\tau^{\mathrm{can}}) + e_p(\widehat{m}_\tau^\kappa, m_\tau^{\mathrm{can}}) + 0.5e_R(\widehat{m}_\tau^\kappa, m_\tau^{\mathrm{can}})\Big] \\ &+ \mathcal{L}_{\mathrm{def}} + 0.01\sum_{\tau=2}^{H}\|u_\tau^{\mathrm{PD}} - u_{\tau-1}^{\mathrm{PD}}\|_2^2 + \mathcal{L}_{\mathrm{terminal}}, \\ \mathcal{L}_{\mathrm{def}} = &\sum_{\tau=1}^{H}\Big[0.1e_q(\widetilde{m}_\tau, m_\tau^{\mathrm{can}}) + 0.1e_p(\widetilde{m}_\tau, m_\tau^{\mathrm{can}}) + 0.05e_R(\widetilde{m}_\tau, m_\tau^{\mathrm{can}})\Big], \\ \mathcal{L}_{\mathrm{terminal}} = &\, 5e_q(\widehat{m}_H^\kappa, m_H^{\mathrm{can}}) + 5e_p(\widehat{m}_H^\kappa, m_H^{\mathrm{can}}). \end{aligned} \tag{C.4}$$

The tracking terms compare realized motion with the fixed canonical target. The deformation terms instead compare the candidate reference with that target, penalizing unnecessary changes to the commands. The action-rate term uses $u_\tau^{\mathrm{PD}} \in \mathbb{R}^{29}$, the low-level PD position targets produced during rollout for the 29 body joints, in radians. Hand targets are excluded. This term sums squared changes across joints and has zero contribution at the first horizon step. It does not measure changes between motion tokens or between the normalized search vectors.

All running terms use uniform time weights of one. The horizon reduction is a sum, while the joint and link reductions in Eq. C.3 are averages. The terminal penalties add weight to joint and link-position errors at the last step; no terminal orientation term is included in this objective.

Control Command Generation uses $N = 80$ candidates per CEM iteration, $I = 8$ iterations per replan, and $E = 9$ elites. The planning horizon is $H = 40$ steps, or 0.80 s, and the retained prefix is $\Delta = 20$ steps, or 0.40 s. After executing that prefix in simulation, the next search continues from its resulting state against the subsequent canonical reference window. Candidate rollouts with non-finite joint, link, or action values receive the failure cost $10^{12}$. The objective otherwise scores the realized motion through the stated tracking terms; it does not add a separate fall or contact heuristic. These settings apply to the second optimization stage for both evaluated deployment controllers.

At the end of a sequence, the reference window is padded by repeating its last frame to retain the full planning horizon. If $n_{\mathrm{rem}}$ control steps remain, only $\min(\Delta, n_{\mathrm{rem}})$ steps are executed.

**Table C.2** Dynamics randomization for Control Command Generation. Multipliers scale the corresponding nominal parameters and remain fixed within each scenario rollout. Velocity offsets are additive and sampled independently for each component; $\mathcal{N}(0, \sigma^2)$ denotes a zero-mean Gaussian distribution with standard deviation $\sigma$.

| Parameter | Distribution | Unit |
|---|---|---|
| Contact friction multiplier | $\mathcal{U}(0.8, 1.2)$ | – |
| Joint armature multiplier | $\mathcal{U}(0.8, 1.2)$ | – |
| Joint friction loss multiplier | $\mathcal{U}(0.8, 1.2)$ | – |
| Actuation strength multiplier | $\mathcal{U}(0.9, 1.1)$ | – |
| Root linear velocity offset | $\mathcal{N}(0, 0.02^2)$ | m/s |
| Root angular velocity offset | $\mathcal{N}(0, 0.05^2)$ | rad/s |
| Joint velocity offset | $\mathcal{N}(0, 0.05^2)$ | rad/s |

*Dynamics randomization during command optimization.* Following (Xie et al., 2026a), we evaluate each command candidate in four randomized physics scenarios during Control Command Generation. At each replanning update, we sample dynamics multipliers and initial-velocity offsets from Table C.2 to construct the four scenarios. The same scenario set is shared by all candidates and CEM iterations within that update; a new set is sampled at the next update. Each scenario rollout starts from the current execution state with the sampled velocity offsets added. The candidate reference and canonical target are shared across scenarios.

We evaluate Eq. C.4 separately in each scenario, using its realized motion and PD targets. For candidates with finite costs in all four scenarios, the CEM score is the arithmetic mean of the four scenario costs. The failure cost specified above is applied at the candidate level: if any scenario produces a non-finite cost, the candidate score is set to $10^{12}$. The count $N = 80$ denotes distinct command candidates, each evaluated in all four scenarios.

After optimization, the selected prefix is executed under nominal simulation dynamics. This execution provides the recorded body states and the starting state for the next planning update. SONIC and HEFT use the same randomization distributions, scenario count, resampling rule, and cost aggregation.

## C.3 SONIC and HEFT Body-Action Labels

SONIC's motion encoder maps the optimized reference to a 64-dimensional motion token (Luo et al., 2026). This token is recorded as the body-action label associated with the selected reference, whereas rollout body-joint angles describe the resulting motion.

HEFT uses 29 absolute body-joint targets together with a nine-dimensional pelvis code (Liu et al., 2026a). Let $p_t \in \mathbb{R}^3$ and $(\alpha_t, \beta_t, \psi_t)$ denote the optimized pelvis position and roll, pitch, and yaw, with position expressed in the world frame. Let $R_z(\psi)$ be the rotation about the world $z$ axis that maps vectors from the yaw-aligned frame into the world frame. Define

$$\begin{aligned} c_t &= \big(\sin\alpha_t, \cos\alpha_t - 1, \sin\beta_t, \cos\beta_t - 1, \delta\psi_t, \delta p_t^\top, p_{t,z}\big)^\top, \\ \delta\psi_t &= \operatorname{wrap}(\psi_{t+1} - \psi_t), \qquad \delta p_t = R_z(\psi_t)^\top (p_{t+1} - p_t). \end{aligned} \tag{C.5}$$

Here wrap returns the principal signed angle modulo $2\pi$. The row combines the current pose's roll, pitch, and absolute height with the transition to the next pose. The translation increment is expressed in the current yaw frame and contains three coordinates. A valid row therefore needs a successor pelvis pose; this is part of the motion-code definition rather than an extra body degree of freedom. Encoding $L + 1$ consecutive pelvis poses produces $L$ code rows; the final input pose supplies the successor needed by the last row.

During decoding, planar position and heading accumulate from an anchor $(\bar{p}_{0,xy}, \bar{\psi}_0)$. Let $L$ be the number of code rows and use local row indices $i = 0, \ldots, L - 1$, independent of the episode time index $t$. With code

components numbered from one, set $\delta\psi_i = c_{i,5}$ and $\delta p_i = (c_{i,6}, c_{i,7}, c_{i,8})^\top$. Then

$$\begin{aligned} \bar{\psi}_i &= \bar{\psi}_0 + \sum_{j=0}^{i-1} \delta\psi_j, \\ \bar{p}_{i,xy} &= \bar{p}_{0,xy} + \sum_{j=0}^{i-1} \big[R_z(\bar{\psi}_j)\delta p_j\big]_{xy}, \qquad \bar{p}_{i,z} = c_{i,9}, \\ \bar{\alpha}_i &= \operatorname{atan2}(c_{i,1}, 1 + c_{i,2}), \qquad \bar{\beta}_i = \operatorname{atan2}(c_{i,3}, 1 + c_{i,4}). \end{aligned} \tag{C.6}$$

Empty sums are zero, and code components are numbered from one. Absolute height directly defines the decoded vertical position; it is not obtained by cumulatively adding the vertical increment. The final row's transition leads beyond the returned sequence of current poses. This explains how the code can contain both vertical displacement and absolute height without integrating both into the same position. The decoded position $\bar{p}_i$ and roll–pitch–yaw angles $(\bar{\alpha}_i, \bar{\beta}_i, \bar{\psi}_i)$ specify the pelvis pose represented by code row $i$. Together with the corresponding 29 absolute body-joint targets, they recover the physical body reference encoded by the HEFT body-action label. HEFT uses pelvis-and-joint references together with the robot state to generate tracking actions, which are mapped to body-joint PD targets. The barred quantities describe reference poses, while the realized motion is obtained from the controller rollout.

## C.4 Hand Targets and Recorded State

Both body-action formats append 12 BrainCo hand-joint targets, six per hand, yielding 76-dimensional SONIC actions and 50-dimensional HEFT actions. The hand command is constructed from the retargeted robot hand geometry. Following EgoHumanoid (Shi et al., 2026), the mean finger curvature is thresholded separately for each hand to distinguish grasp from release. For grasp, the command uses the calibrated BrainCo grasp pose to encourage active closure. For release, it uses the current retargeted joint angles. These targets are delivered separately to hand control. This construction specifies the closure command without interpreting a retargeted geometric pose as a measured grasp force.

The policy state has the same 44-dimensional format for both controllers:

$$s_t^\kappa = \left(b_t^{\mathrm{roll},\kappa},\ (R_t^{\mathrm{base},\kappa})^\top g^W,\ f_t^{\mathrm{ret}}\right). \tag{C.7}$$

Here $b_t^{\mathrm{roll},\kappa} \in \mathbb{R}^{29}$ contains rollout body-joint angles and $f_t^{\mathrm{ret}} \in \mathbb{R}^{12}$ contains continuous retargeted hand angles, both in radians. The rotation $R_t^{\mathrm{base},\kappa} \in SO(3)$ maps base-frame vectors to the world frame, and $g^W = (0, 0, -1)^\top$ is the dimensionless unit gravity direction. Thus $(R_t^{\mathrm{base},\kappa})^\top g^W$ gives the three-dimensional projected-gravity observation in the rollout base frame. The state records the estimated hand geometry even when a grasp action requests the calibrated closure target. Replacing the state hand angles with that target would change the meaning of the observation. Body-state values depend on the selected controller's rollout, although the state format and retained hand estimates are shared.

## C.5 Episode Export and Observation Reuse

The procedure below summarizes the conversion from a canonical episode to controller-specific supervision.

**Controller-specific episode export**

1. Read the canonical motion, synthesized images, retargeted hand trajectory, and instruction on their common temporal index.
2. For each receding-horizon window, decode CEM residuals into candidate body references, encode them for controller $\kappa$, simulate, and evaluate Eq. C.4 against the fixed canonical target.
3. Retain the selected prefix of up to 20 steps, shortened when fewer steps remain, and continue from its resulting simulated state. Collect the optimized reference and realized body motion.
4. Record the selected reference as SONIC tokens or HEFT body labels; append the 12 grasp/release hand targets.

**Table D.1** Task goals and the changes defining Robot-OOD. Robot-ID uses the appearance and pose conditions represented in the robot demonstrations.

| Task | Placement goal | Robot-ID to Robot-OOD change |
|---|---|---|
| Cup-to-basket | Place the object in a basket | Green bottle replaced by a brown bottle |
| Parcel-to-shelf | Place the parcel on a shelf | Purple parcel replaced by a white parcel |
| Basket-to-shelf | Place the basket on a shelf | Basket handle tilted by approximately 30° |
| Cup-to-drawer | Place the object in a drawer | Green bottle replaced by a brown bottle |

5. Assemble the 44-dimensional state from rollout body joints, projected gravity, and retargeted hand angles.
6. Associate these states and actions with the retained canonical images and instruction to obtain the controller-specific training episode.

Canonical-image reuse follows the fixed reference trajectory, while the recorded body state follows the selected controller's realized motion. Their correspondence is therefore approximate when tracking is imperfect. The wrist and camera deviations in TABLE 4 measure this residual mismatch for the evaluated controllers, and TABLE 1 evaluates the utility of the resulting episodes for co-training. Changing from SONIC to HEFT repeats command optimization, encoding, and state recording while retaining canonical motion generation and view synthesis. The reuse rests on tracking the shared reference sufficiently well for the evaluated setting; it does not assert identical camera poses across controller rollouts.

# D Experimental Protocol and Metric Definitions

This appendix specifies the data composition, controlled comparisons, and interpretation of the evaluation metrics used in Sec. 4.1. The task-success results remain in TABLES 1–3; the motion and rollout errors remain in TABLE 4.

## D.1 Demonstration Sources and Evaluation Conditions

The experimental platform is a Unitree G1 with BrainCo dexterous hands. For each task and deployment controller, the robot dataset contains 50 demonstrations collected through the corresponding SONIC or HEFT PICO teleoperation pipeline. The human pool contains 100 demonstrations per task, recorded with a chest-mounted iPhone using MobileEgo Anywhere (Palanisamy et al., 2026). All 400 human clips include phone-recorded depth, and both demonstration sources use 50 Hz sequences. The same human canonical records supply the motion, synthesized observations, and instructions for both controllers; command generation and state recording produce the controller-specific training fields.

TABLE D.1 specifies the task goals and condition changes. Robot-ID is defined relative to object appearance and pose conditions in the robot demonstrations. Robot-OOD changes these conditions beyond that robot dataset. Each task's human pool contains 80 shifted and 20 robot-matched clips, so the shifted test conditions are represented in the human training pool. This evaluates whether converted human data supplies coverage absent from the robot demonstrations. It does not test conditions excluded from both training sources. The 80:20 composition describes the complete human pool; the independently sampled smaller budgets are not specified as having that same proportion.

## D.2 Training Budgets and Success-Rate Aggregation

For each controller configuration, GR00T N1.7 (Bjorck et al., 2025) is trained jointly on all four tasks for 20,000 iterations with batch size 160. Within each task, robot and converted-human sources are sampled in proportion to their episode counts; other sampling settings follow GR00T defaults.

The scaling study combines 50 fixed robot demonstrations with 0, 25, 50, 75, or 100 converted-human demonstrations per task. Human subsets are sampled randomly and independently at each budget without

nesting. Policy architecture and training budget remain fixed. All component ablations use SONIC with 50 robot and 100 converted-human demonstrations per task.

Each configuration is trained once and evaluated in 20 real-robot trials per task under each of the Robot-ID and Robot-OOD conditions. A trial succeeds when the placement is completed without a fall or unrecovered manipulation failure. We report the percentage of successful trials for each task and the unweighted mean across the four tasks, separately for each condition.

## D.3 Controlled Component Comparisons

TABLE D.2 separates the observation variants from the control-command variants. In the observation study, the demonstration set, state–action supervision, policy architecture, training budget, and real-robot evaluation protocol are fixed. Only the human-derived observation construction changes. Fixed offset adapts the view-alignment strategy of EgoHumanoid (Shi et al., 2026) to chest-mounted capture, using a 0.10 m vertical shift and ±0.05 m training-time perturbations. Both reprojection-only variants include hole inpainting.

**Table D.2** Definitions of the component variants. All comparisons use SONIC and the full human-data budget. The table describes changed components; quantitative policy results are in TABLES 2 and 3.

| Variant | Observation or command construction |
|---|---|
| **Observation alignment** | |
| No alignment | Original human images. |
| Fixed offset | Scene reprojection with a fixed vertical camera shift, training-time perturbations, and hole inpainting. |
| Motion-based reprojection | Scene reprojection and hole inpainting using camera poses derived from the canonical robot motion. |
| Full alignment | Motion-derived onboard views, additionally removing human arms and rendering and compositing the target robot. |
| **Control Command Generation** | |
| Without SCO | Canonical whole-body motion used directly as the tracking controller's reference; command optimization omitted only at this stage. |
| With SCO | Controller-specific commands optimized to track the same canonical motion. |

Full alignment changes human-arm removal and robot rendering/compositing together relative to motion-based reprojection. Consequently, this comparison does not separately identify the effect of each image-processing operation. The main interpretation concerns the complete alignment pipeline relative to original human observations and the adapted fixed-offset baseline. The intermediate variant describes a partial pipeline under the same training setup.

In the command-generation study, both variants retain Whole-Body Motion Generation and Robot Onboard View Generation. They use identical canonical trajectories, observations, demonstrations, and policy-training settings. The optimized commands are evaluated through the deployment controller, and body-state features are recorded from the resulting rollouts. Changing command generation can therefore change both action labels and body-state features. This is a comparison of state–action supervision as a whole, not an action-only intervention with all state inputs held fixed. These component comparisons use SONIC; the data-scaling study evaluates reuse of the complete converted records under both SONIC and HEFT.

## D.4 Motion References, Units, and Refinement Replay

The blocks of TABLE 4 use different references. Whole-body motion-generation errors compare the generated robot motion with the source-derived motion references. Controller-rollout errors compare the realized motion with the fixed canonical trajectory. The former measures source-reference fidelity, while the latter measures how closely a particular controller realizes the canonical motion. Direct denotes references without command optimization at the evaluated stage, SCO denotes CEM-optimized commands, and SCO+IK adds Mink arm refinement during whole-body motion generation.

Position errors are reported in centimeters, and orientation errors and joint-angle RMSE are reported in radians. The motion-generation block reports wrist and torso position and orientation errors. The controller-rollout blocks report wrist and camera position and orientation. The smallest reported values are identified separately for each target and error type. Improved position tracking does not necessarily imply improved orientation tracking: SONIC's camera-orientation error is 0.0784 rad for Direct and 0.0808 rad for SCO. Dashes denote unreported metrics or variants.

Motion-generation entries average aggregate errors from LAFAN (Harvey et al., 2020) and the 400 egocentric training clips. LAFAN provides an auxiliary check on highly dynamic motion, while the egocentric clips connect the diagnostics to the converted policy-training data. Both controllers share evaluation data. The reported aggregate entries do not provide separate results for the two data sources.

TABLE D.3 expands the before/after-Mink replay diagnostics described in Sec. 4.5. The same SONIC tracker replays the pre-refinement and refined trajectories on both data sources. For each replay, tracking errors are measured against the corresponding input reference trajectory. The diagnostics include pelvis-relative non-arm and wrist-position errors, torso-orientation error, and non-arm joint-angle RMSE. These replay errors measure controller trackability and are distinct from motion-generation errors measured against the source-derived references. Although refinement preserves the non-arm reference trajectory, changes to the arm reference can affect the realized whole-body motion and hence the non-arm replay errors. The small changes indicate comparable SONIC trackability on the evaluated trajectories; they do not constitute a downstream policy-success ablation of Mink refinement.

**Table D.3** SONIC replay before and after Mink arm refinement on LAFAN and the 400 egocentric clips. Each replay is evaluated against its corresponding input reference trajectory.

| Replay metric | Unit | Before refinement | After refinement |
|---|---|---|---|
| Pelvis-relative non-arm position error | cm | 2.60 | 2.73 |
| Pelvis-relative wrist-position error | cm | 1.59 | 1.75 |
| Torso-orientation error | rad | 0.0588 | 0.0579 |
| Non-arm joint RMSE | rad | 0.0750 | 0.0784 |

## E Qualitative Demonstration Timelines

Each timeline spans one complete source clip through eight selected keyframes, including its first and last frames. Read left to right, then top to bottom; timestamps refer to the original 50 Hz overlay video. Each frame retains the human view (upper left), synthesized robot view (upper right), and offline MuJoCo kinematic replay (main panel). These illustrate converted demonstrations; they are not additional real-robot evaluation trials.

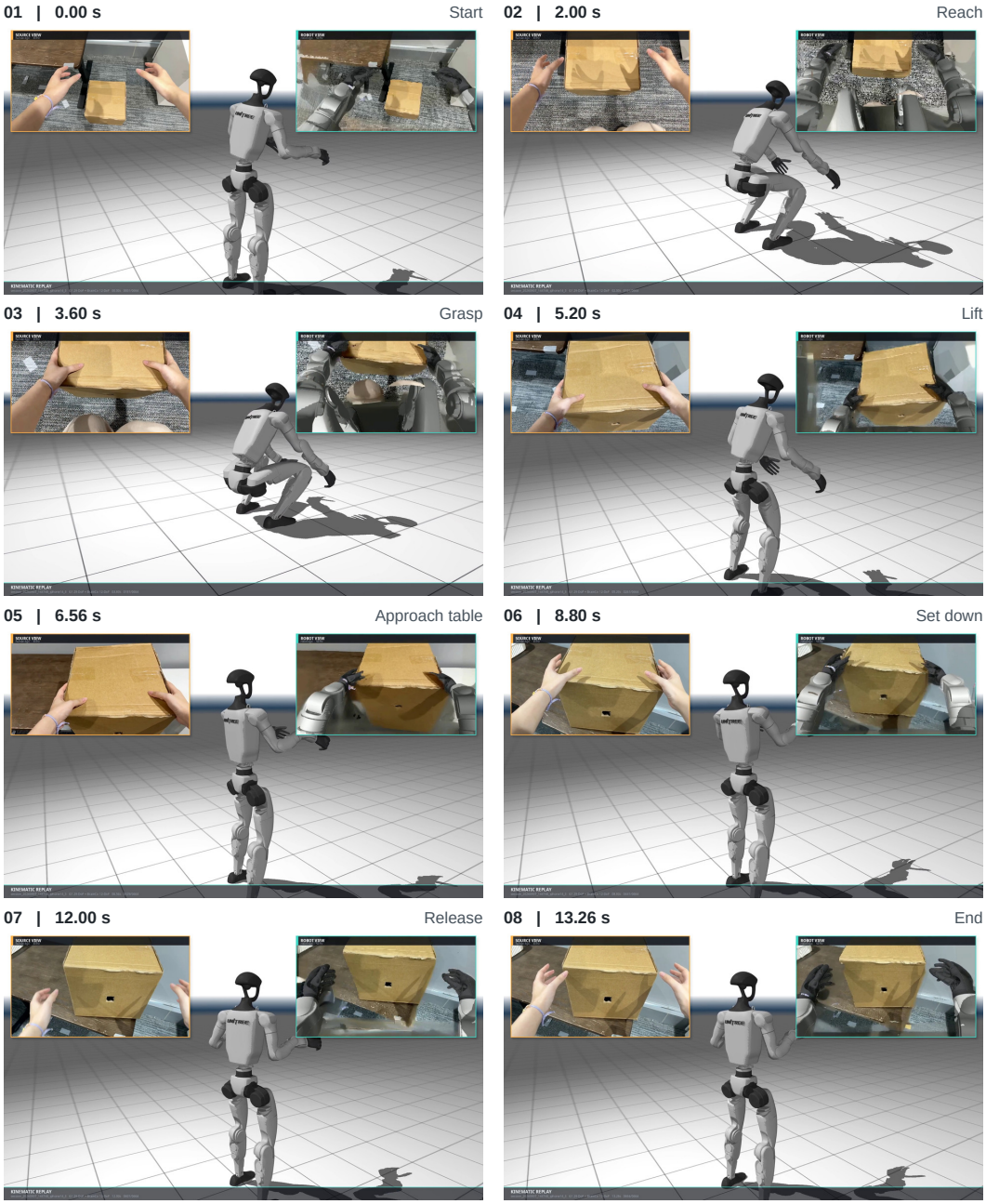



**Figure E.1 Complete-clip timeline: bimanual box transfer.** Selected moments from a 13.28 s clip show reaching toward a box, lifting and carrying it, and setting it on a table before withdrawing the hands. Frames 1–8 preserve chronological order and the paired human, robot-view, and whole-body-motion displays.

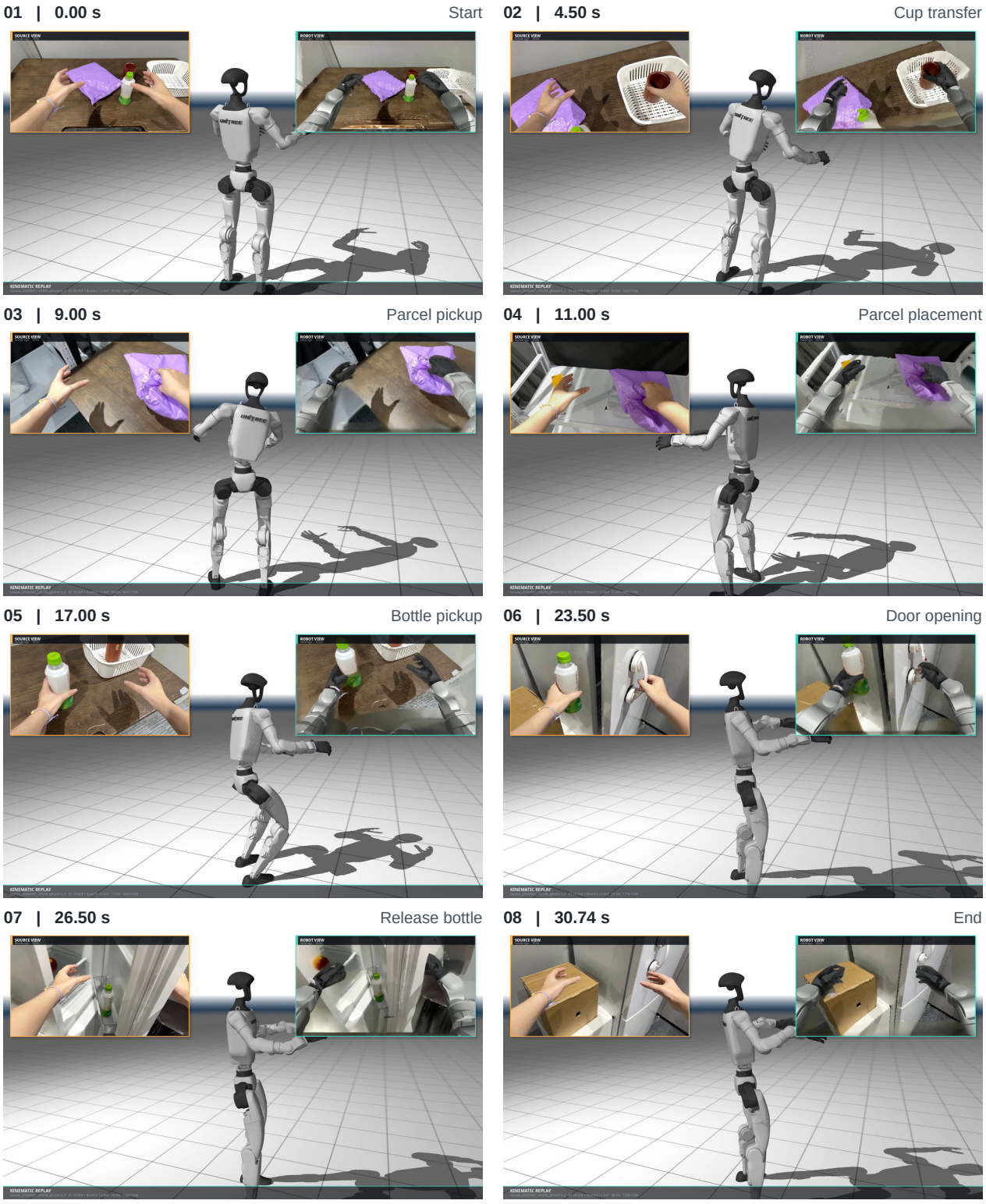



**Figure E.2 Complete-clip timeline: sequential object manipulation.** Eight selected moments span a 30.76 s clip: a cup is moved toward a basket, a parcel is transferred to a shelf, and a bottle is carried to a refrigerator and placed inside before the door is closed. The first and last video frames are included. Time advances left to right and then top to bottom; the three synchronized displays have the same meanings as in Fig. E.1.

# F Additional Task Demonstrations and Deployments

Figures F.1–F.3 complement Fig. 4 with the other three evaluated tasks. Each figure follows the same row order: Robot Data in the Robot-ID condition, converted KODA Data, and successful SONIC deployment in the Robot-OOD condition. Four keyframes show the action progression within each row. Timestamps are relative to the corresponding source clip; different rows are independent recordings and are not temporally synchronized. The KODA row retains the human view, synthesized robot view, and offline kinematic replay described in Appendix E. Bystanders are cropped out or pixelated.

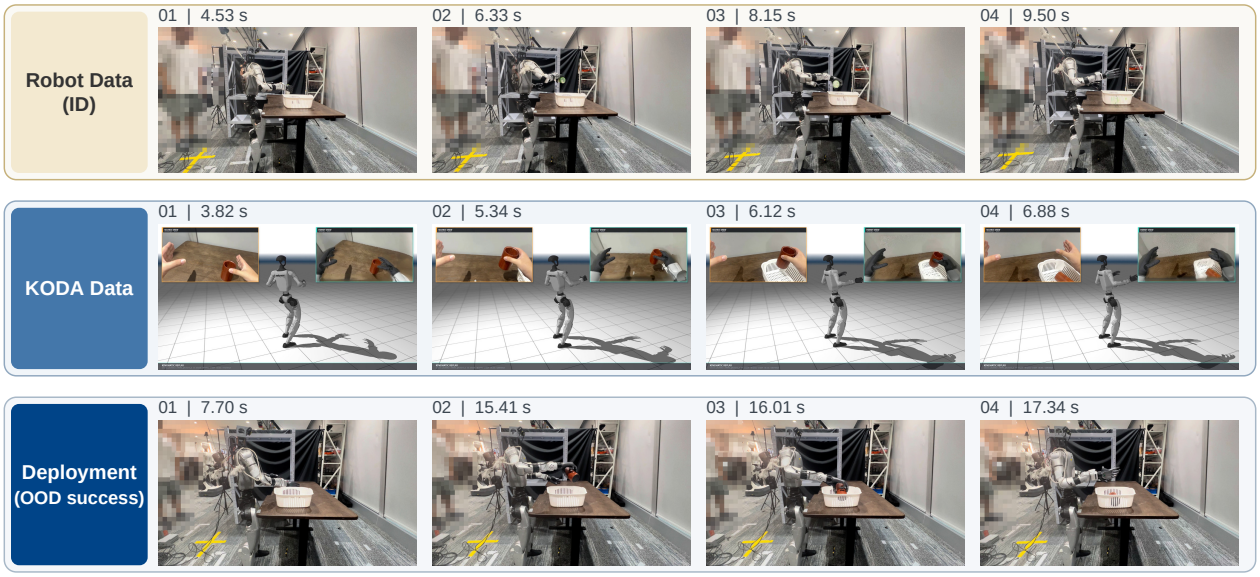



**Figure F.1 Cup-to-basket.** Robot Data shows the green bottle used in Robot-ID. KODA Data covers the brown-bottle condition, and the deployment row shows a successful Robot-OOD placement into the basket. Each row progresses from grasping through transfer to placement.

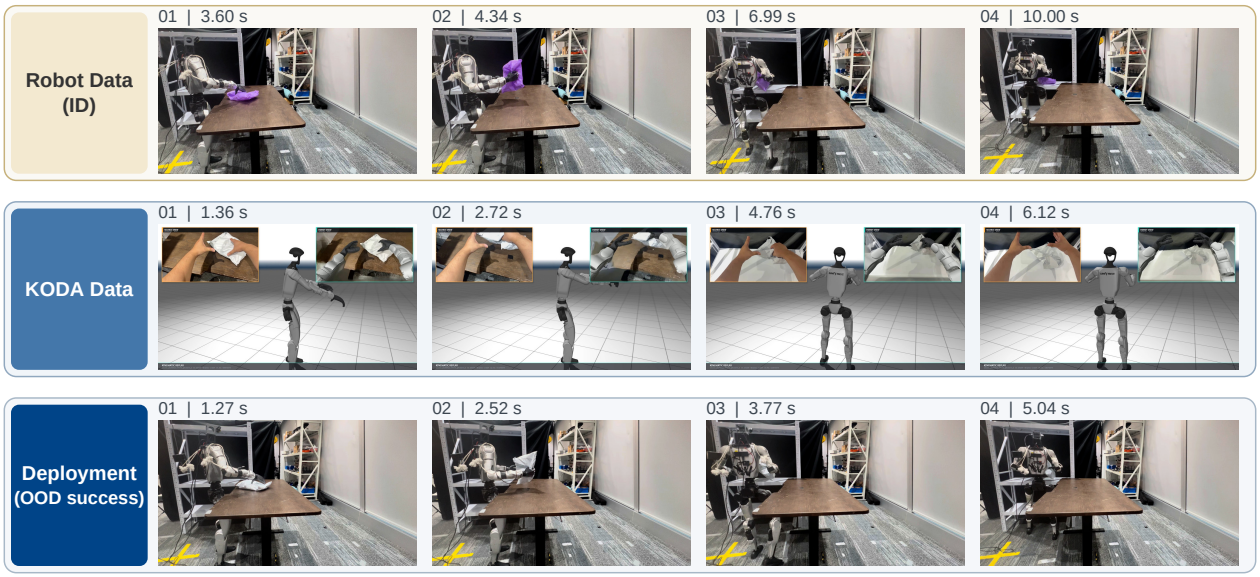



**Figure F.2 Parcel-to-shelf.** The purple parcel in Robot Data is replaced by a white parcel in the KODA and Robot-OOD deployment examples. The keyframes show reaching, lifting and transferring the parcel, followed by placement on the shelf.

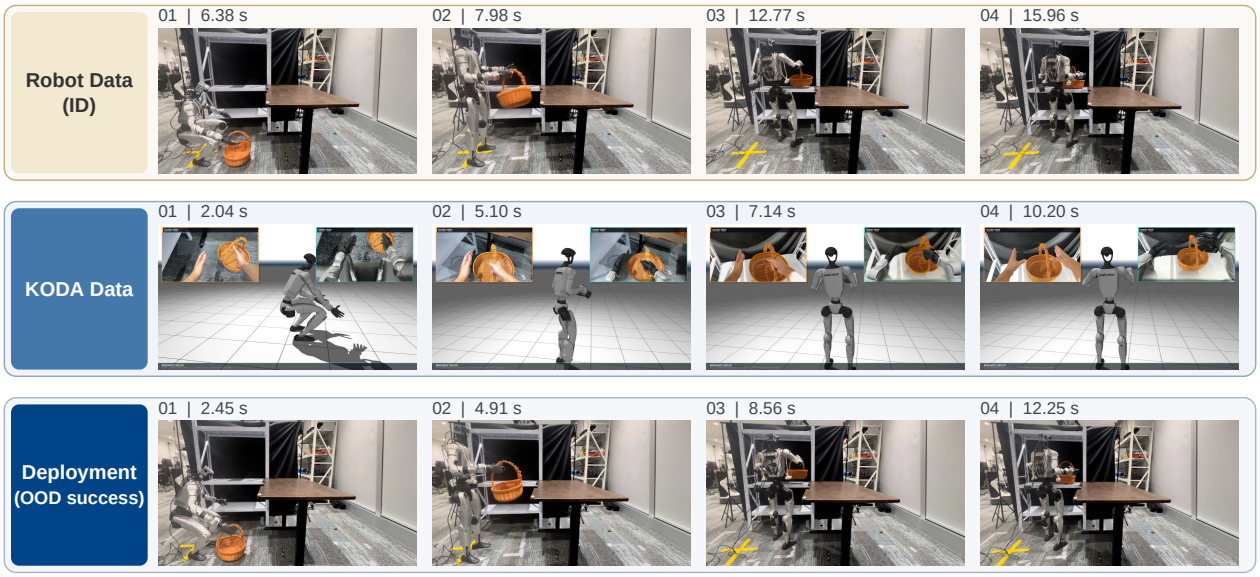



**Figure F.3 Basket-to-shelf.** The Robot-OOD change is an approximately 30° tilt of the basket handle relative to the Robot-ID condition. KODA Data supplies a corresponding converted demonstration; the deployment row shows grasping the handle, lifting the basket, and placing it on the shelf.